Natural Language Processing
1038141

1.9 - Syntax

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**credits are reported in the last slide

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1.9 - Syntax

- Syntax
- Formal Grammars, Context-Free Grammars, Chomsky Normal Form, Constituent Grammars vs. Dependency Grammars, Ambiguity, Treebanks Syntactic Parsing, Cocke-Kasami-Younger Algorithm, Early Algorithm, Probabilistic CFG, Probabilistic CKY Algorithm
- Exercises
- Q&A

Milestones in NLP

today topic is fundamental for the three areas.

rule-based systems

statistical classical machine learning models

deep learning models

Syntax

definition:

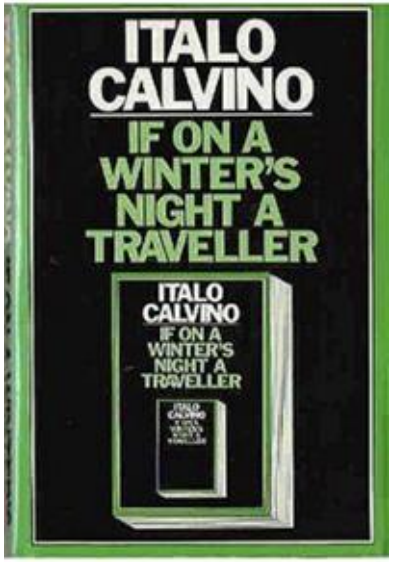
Syntax is the branch of linguistics that studies the way words are arranged together in sentences.

Syntax

Consider the following book title:

"If on a winter's night a traveler"

and the following translations:



- *Se una notte d'inverno un viaggiatore*
- **Se notte una d'inverno un viaggiatore*
- *Una notte se d'inverno un viaggiatore*
- **Se un notte d'inverno una viaggiatore*
- *Se una notte un viaggiatore d'inverno*
- *Se un viaggiatore d'inverno una notte*
- **Se un una notte viaggiatore d'inverno*
- **Se un una d'notte viaggiatore inverno*
- *~Se un inverno d'notte un viaggiatore*
- *Se d'inverno un viaggiatore una notte*
- *Se d'inverno una notte un viaggiatore*

Syntax

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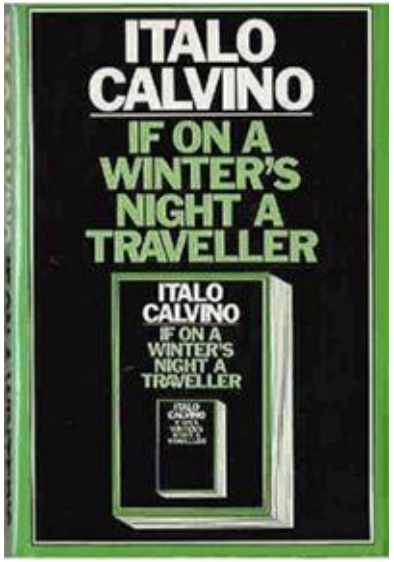

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- *Se d'inverno una notte un viaggiatore*

- some are good translations
- others are not syntactically correct
- part of the problem is related to the **order** of words

Syntax

The “word order” problem has been partially addressed with:

- Regular languages and regular expressions
- N-gram models: Computing probabilities for word sequences
- Part-of-speech classes: equivalence classes for words

We move to …

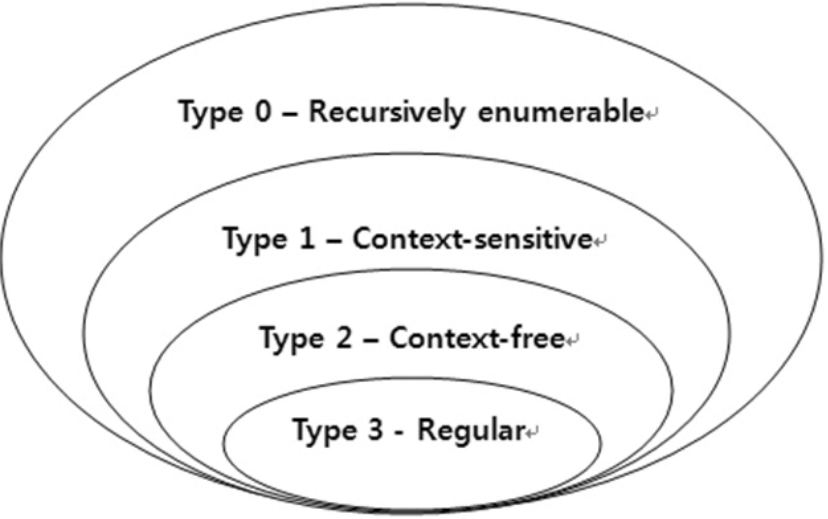
We move to …

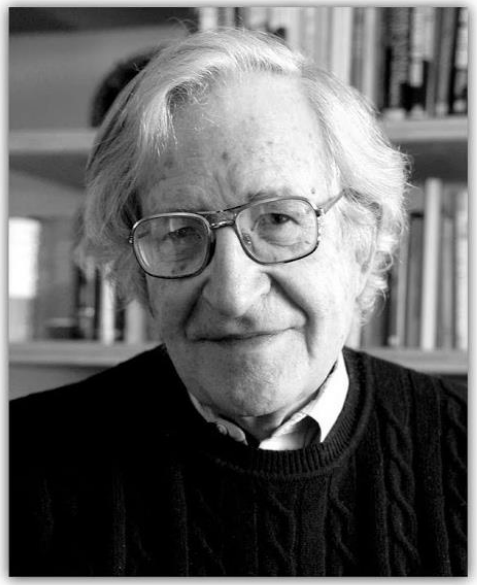
formal grammars!

Formal Grammars

Grammars are able to determine the correct syntax of a language.

We like **Grammars** but we love **Formal Grammars.**





Language	Automaton	Grammar	Recognition
Recursively Enumerable Languages	Turing Machine	Unrestricted $Baa \rightarrow A$	Undecidable ?
Context-Sensitive Languages	Linear Bounded	Context Sensitive $A\,t \rightarrow aA$	Exponential?
Context-Free Languages	Pushdown Stack	Context Free $S \rightarrow gS\,c$	Polynomial
Regular Languages	Finite-State Automaton	Regular $A \rightarrow cA$	Linear

Context-free Grammars

A context-free grammar (CFG) or phrase-structure grammar is a formal grammar defined as a 4-tuple:

$$G = (N, T, P, S)$$

where:

- N is the set of nonterminal symbols (phrases or clauses)
- T is the set of terminal symbols (lexicon)
- P is the set of productions (rules), $P \subseteq N \times (N \cup T)^*$
- S is the start symbol such that $S \in N, \exists (S, \beta) \in P$

Example of CFG

$$G = (N, T, P, S)$$

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N = { S , NP, Nom, Det, Noun }
T= { a, the, winter, night }
P = { S → NP,
NP → Det Nom,
Nom → Noun | Nom Noun,
Det → a | the,
Noun → winter,
Noun → night }
S

CFGs

N = { S , NP, Nom, Det, Noun }
T= { a, the, winter, night }
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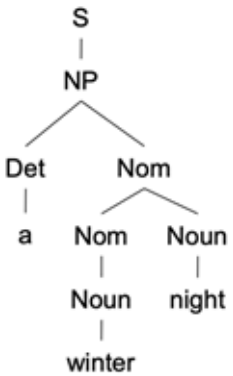
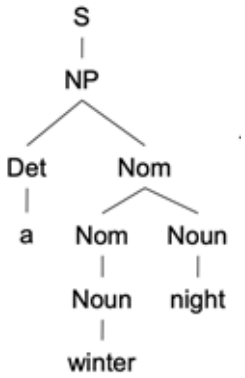
Generating sentences

Analysing sentence structures

derivations

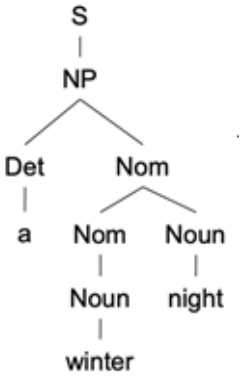


all possible derivations



"a winter night"

Parse tree



"a winter night"

L(G)

definition:
L(G) is the set of strings of terminals w for which there is a sequence of productions (derivation) that yield w from S.

formally:

$$L(G) = \{ w \in T^* : S \Rightarrow^* w \}$$

L(G) is the language of the grammar G
L(G) is a context-free language if G is context-free
G is a CFG if is defined as:

$$G = (N, T, P, S)$$

where:

- N is the set of nonterminal symbols (phrases or clauses)
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CFG in Normal Forms

a CFG is in Chomsky Normal Form (CNF) if:

- each production is in one of the following form:

  $A \rightarrow B\ C$

  $A \rightarrow a$
- the production of ε is not allowed (ε-free)

CNF grammars generate binary branching

Any CFG grammar can be converted into a weakly equivalent CNF grammar:

Convert rules of the kind: $A \rightarrow B\ C\ D$ into two productions:

- $A \rightarrow B\ X$
- $X \rightarrow C\ D$

Dependency Grammars

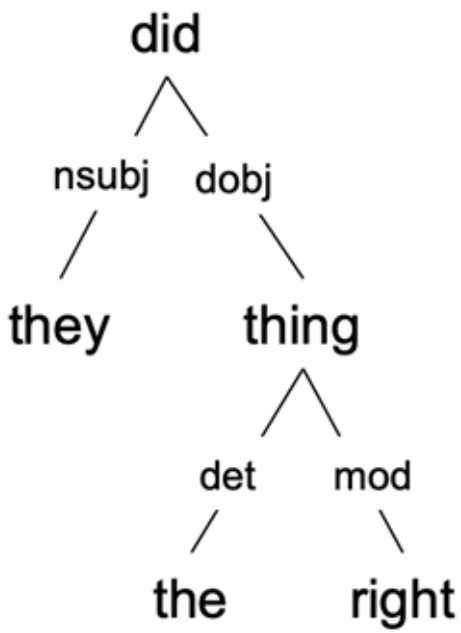
instead of defining CFGs where N as a set of constituents (e.g. NP etc…) which enable the definition of phrase-structure rules;

we define CFGs, where:
productions express lexical, syntactic, or semantic relations between words, called dependencies

each production is labelled with a dependency type

Example of a typed dependency parse tree

- *"They did the right thing"*



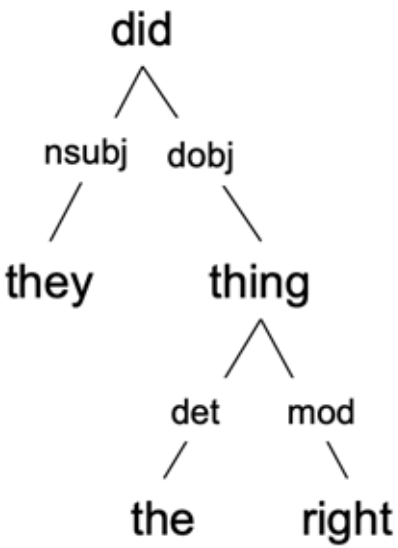
Argument Dependencies	Description
nsubj	They → did
dobj	did → thing
iobj	gave → me
pobj	to → me
Modifier Dependencies	**Description**
tmod	studied → yesterday
det	the → thing
prep	on → shelf

nsubj: nominal subject
dobj: direct object
iobj: indirect object
pobj: object of preposition
tmod: temporal modifier
det: determiner
prep: prepositional modifier
…

https://downloads.cs.stanford.edu/nlp/software/dependencies_manual.pdf

- No non-terminal or phrasal nodes!
- derivation links are labeled (a.k.a. typed dependency parse)

Advantages of Dependency Grammars



the structures are word order-free:

- better for languages like the Italian or Czech in which the order of words is much more flexible;
- a phrase-structure grammar would need separate rules for each possible position in the parse tree.

Treebanks

- CFGs can be used to assign a parse tree to any valid sentence
- We can build a corpus, called treebank, whose sentences are annotated with parse trees
- The most popular project of this kind is the **Penn Treebank**
  - From the Brown, Switchboard, ATIS and Wall Street Journal corpora of English
    - Wall Street Journal: 1.3 million words
    - Brown Corpus: 1 million words
    - Switchboard: 1 million words
  - All tagged with Part-of-Speech & syntactic structure
  – Developed 1988-1994

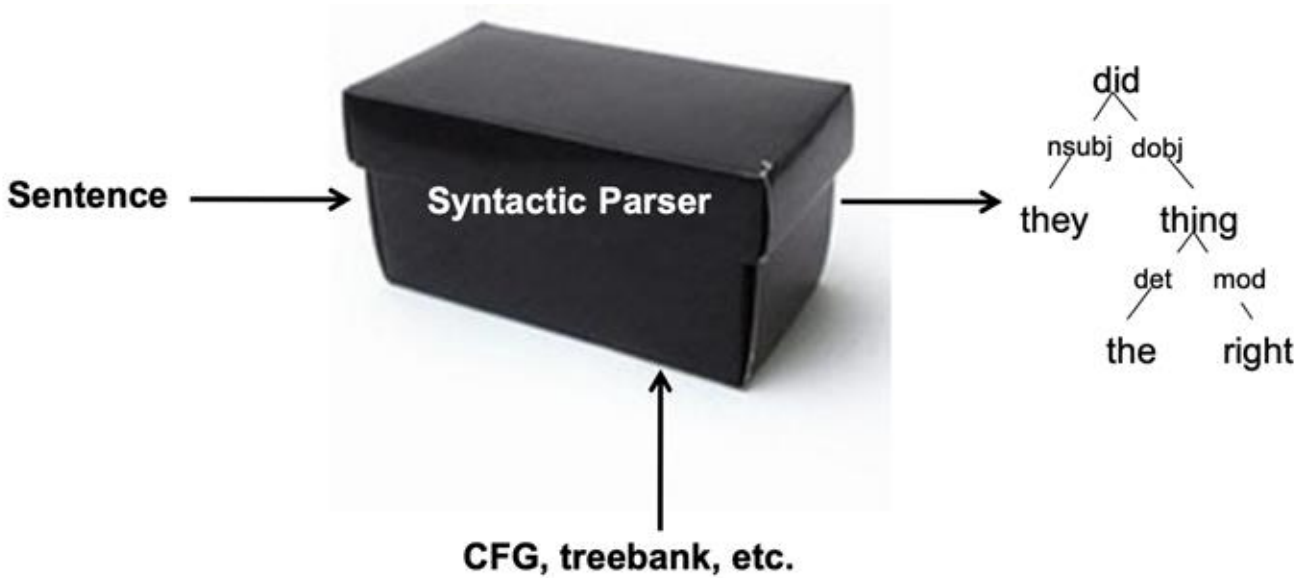
Universal dependencies (https://universaldependencies.org)

A project that provides treebank annotations that:
– are cross-linguistically consistent for many languages
– facilitate:
    multilingual parser development
    cross-lingual learning
    parsing research also across languages

Syntactic Parsing

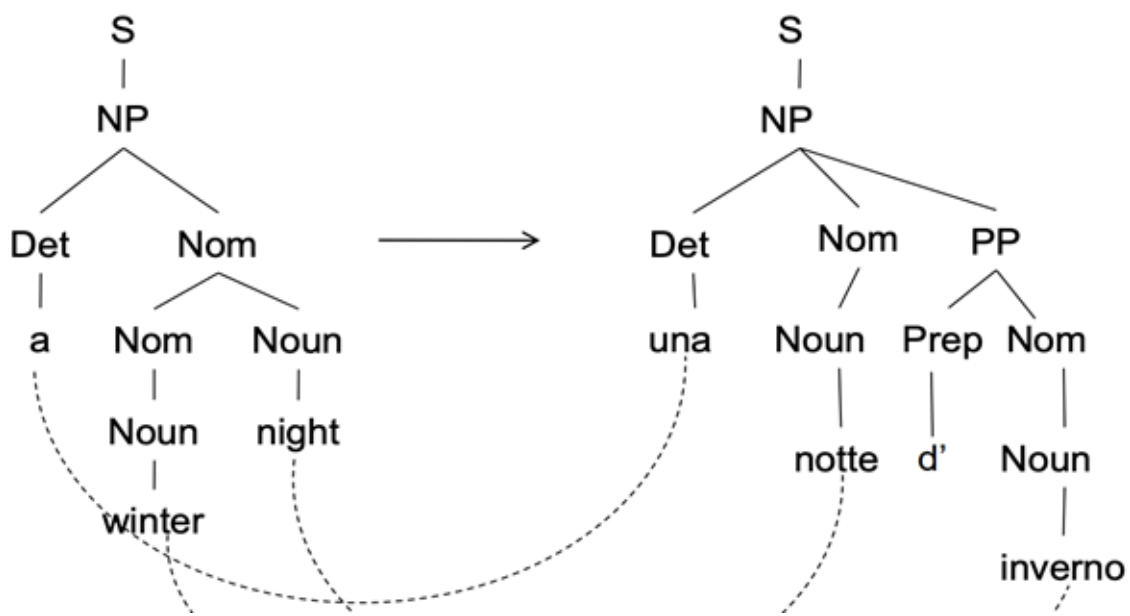
definition:

Syntactic Parsing is the task of recognizing a sentence and assigning a syntactic structure to it.



Why Parsing a Sentence?

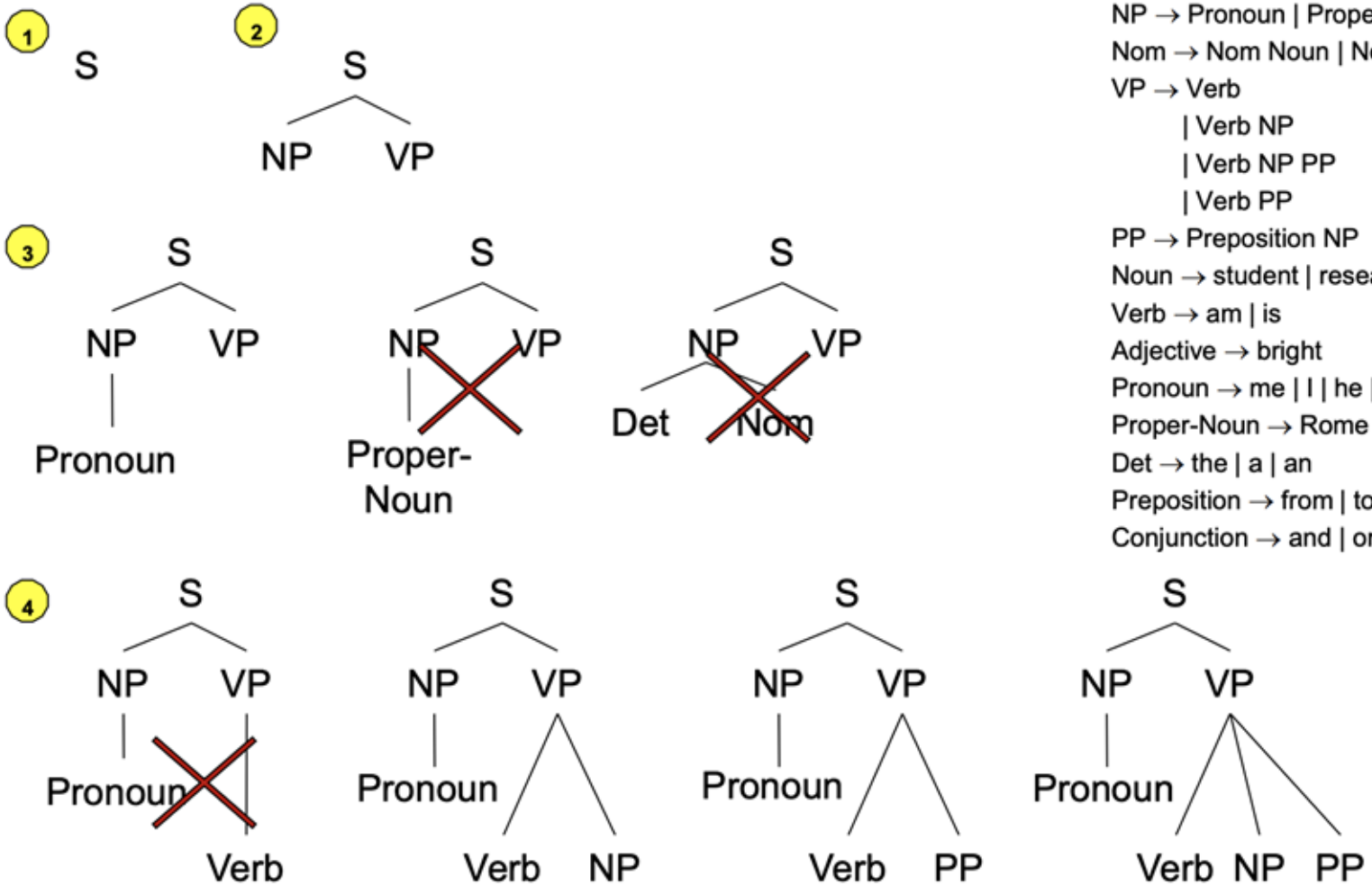
- **Grammar Checking:** dozens of real applications …;
- **Semantic analysis:** to provide a ground for further analysis (e.g., the semantic one);
- **Question Answering:** a Question Answering system needs (at least…): the subject of the sentence (what books), the main verb (write), the by-adjunct (Raymond Queneau), etc;
- **Machine Translation:** to provide a syntactic-coherent ground alignment.



Constituency Parsing is a Search Task

- We search through the space of possible parse trees for the best parse tree for a given sentence
- We have two constraints:
  1. Data constraint: given an input sentence of k words, the parse tree **must have k leaves**
  2. Grammar constraint: the final parse tree **must have one root**
- Depending on the constraint we start from, we have two search strategies:
  - Top-down or goal-directed search
  - Bottom-up or data-directed search

Top-down Parsing: “He is a student in Rome”



S → NP VP
NP → Pronoun | Proper-Noun | Det Nom
Nom → Nom Noun | Noun
VP → Verb
| Verb NP
| Verb NP PP
| Verb PP
PP → Preposition NP
Noun → student | researcher | research
Verb → am | is
Adjective → bright
Pronoun → me | I | he | you | it | him | her
Proper-Noun → Rome | Sapienza
Det → the | a | an
Preposition → from | to | on | in | near | at
Conjunction → and | or | but

Bottom-up Parsing: “He is a student in Rome”

1 He is a student in Rome

2 Pronoun Verb Det Noun Prep Proper-Noun
He is a student in Rome

3 NP Nom NP
Pronoun Verb Det Noun Prep Proper-Noun
He is a student in Rome

4 NP PP NP VP
Nom NP
Pronoun Verb Det Noun Prep Proper-Noun
He is a student in Rome

NP PP NP
Nom NP
Pronoun Verb Det Noun Prep Proper-Noun
He is a student in Rome

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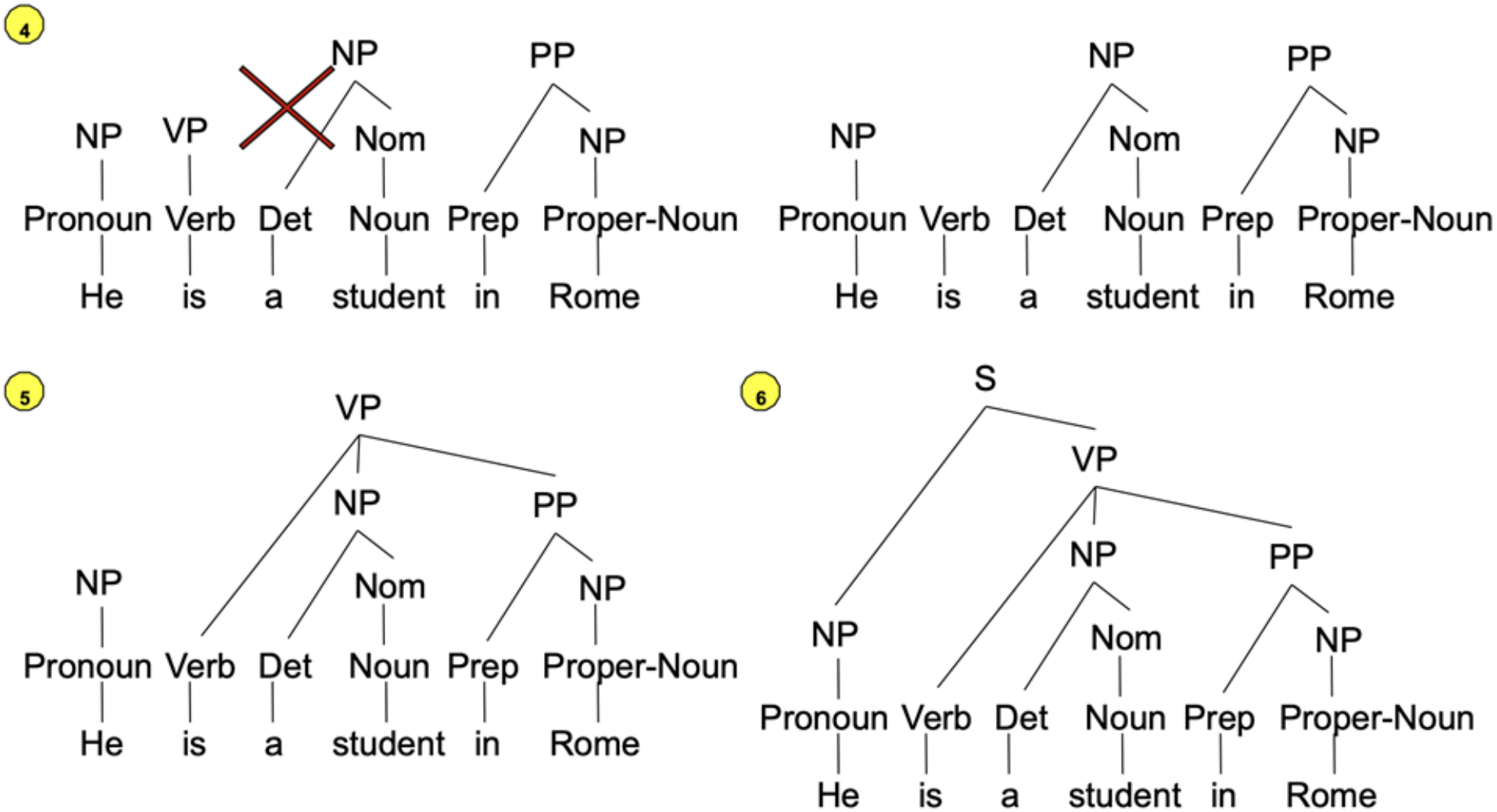
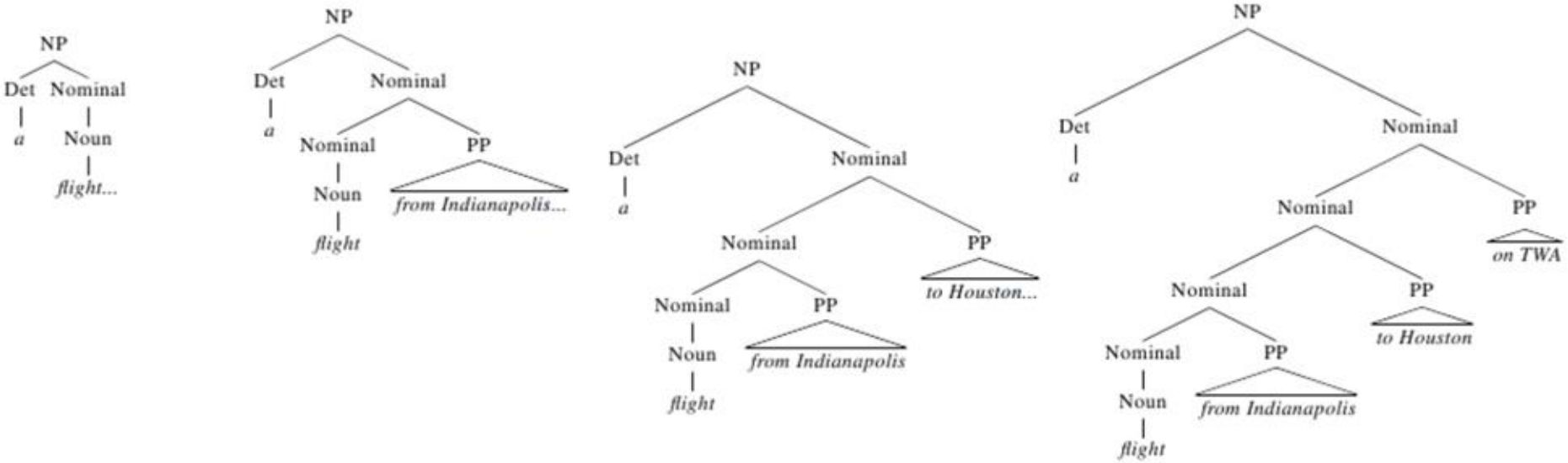


Structural Ambiguity

- A kind of ambiguity which arises in parse trees
- Structural ambiguity is due to the existence of more parse trees for the same sentence
- Choosing the correct parse tree is a matter of syntactic disambiguation
- Even if a sentence is not ambiguous, it can be inefficient to parse because of local ambiguity
  - The sentence is not ambiguous globally, like in “Book that flight”
  - When the parser sees “book” it does not know its POS

Using Backtracking to Parse

- Is a top-down, depth-first, left-to-right backtracking strategy efficient?



- Much parsing work is repeated many times

Dynamic Programming

- Dynamic programming uses memory tables to save solutions to sub-problems
- Tables save subtrees for each constituent of the input sentence
  - No repeated parsing of the same phrase
  - Subtrees are looked up, not parsed again and again!

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Can you tell when we already discussed of Dynamic Programming algorithms in this course?

The Cocke-Kasami-Younger (CKY) Algorithm

- A bottom-up dynamic programming parsing approach
- Takes as input a CFG in Chomsky Normal Form
- Given a sentence of n words, we need an (n+1)x(n+1) matrix
- Cell (i,j) contains the set of non-terminals that produce all the constituents spanning positions from i to j of the input sentence
- The cell that represents the entire sentence is (0,n)
- Main idea: if a non-terminal A is in (i,j), there is a production $A \rightarrow B\ C$, so there must be an intermediate position k with B in (i,k) and C in (k,j)

How to fill in the table?

- Bottom-up
- Given a cell (i,j) all the cells containing its parts are already filled
  - Cells to the left and below

```
function CKY-PARSE(words, grammar) returns table

  for j ← from 1 to LENGTH(words) do
      for all {A | A → words[j] ∈ grammar}
              table[j − 1, j] ← table[j − 1, j] ∪ A
      for i ← from j − 2 down to 0 do
          for k ← i + 1 to j − 1 do
              for all {A | A → BC ∈ grammar and B ∈ table[i,k] and C ∈ table[k, j]}
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- An input sentence is recognized if (0,n) contains S

Exercise

Book the flight through Houston

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```

words

Book	the	flight	trough	Houston
[0,1]	[0,2]	[0,3]	[0,4]	[0,5]
	[1,2]	[1,3]	[1,4]	[1,5]
		[2,3]	[2,4]	[2,5]
			[3,4]	[3,5]
				[4,5]

$S \rightarrow NP\ VP$
$S \rightarrow X1\ VP$
$X1 \rightarrow Aux\ NP$
$S \rightarrow book \mid include \mid prefer$
$S \rightarrow Verb\ NP$
$S \rightarrow X2\ PP$
$S \rightarrow Verb\ PP$
$S \rightarrow VP\ PP$
$NP \rightarrow I \mid she \mid me$
$NP \rightarrow TWA \mid Houston$
$NP \rightarrow Det\ Nominal$
$Nominal \rightarrow book \mid flight \mid meal \mid money$
$Nominal \rightarrow Nominal\ Noun$
$Nominal \rightarrow Nominal\ PP$
$VP \rightarrow book \mid include \mid prefer$
$VP \rightarrow Verb\ NP$
$VP \rightarrow X2\ PP$
$X2 \rightarrow Verb\ NP$
$VP \rightarrow Verb\ PP$
$VP \rightarrow VP\ PP$
$PP \rightarrow Preposition\ NP$

Lexicon

$Det \rightarrow that \mid this \mid the \mid a$
$Noun \rightarrow book \mid flight \mid meal \mid money$
$Verb \rightarrow book \mid include \mid prefer$
$Pronoun \rightarrow I \mid she \mid me$
$Proper\text{-}Noun \rightarrow Houston \mid NWA$
$Aux \rightarrow does$
$Preposition \rightarrow from \mid to \mid on \mid near \mid through$

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j=1

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	[1,2]	[1,3]	[1,4]	[1,5]
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j=1

words

Book	the	flight	trough	Houston
S,VP, Nominal Noun,Verb [0,1]	[0,2]	[0,3]	[0,4]	[0,5]
	[1,2]	[1,3]	[1,4]	[1,5]
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words

Book	*the*	*flight*	*trough*	*Houston*
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$S \rightarrow NP\ VP$
$S \rightarrow X1\ VP$
$X1 \rightarrow Aux\ NP$
$S \rightarrow book \mid include \mid prefer$
$S \rightarrow Verb\ NP$
$S \rightarrow X2\ PP$
$S \rightarrow Verb\ PP$
$S \rightarrow VP\ PP$
$NP \rightarrow I \mid she \mid me$
$NP \rightarrow TWA \mid Houston$
$NP \rightarrow Det\ Nominal$
$Nominal \rightarrow book \mid flight \mid meal \mid money$
$Nominal \rightarrow Nominal\ Noun$
$Nominal \rightarrow Nominal\ PP$
$VP \rightarrow book \mid include \mid prefer$
$VP \rightarrow Verb\ NP$
$VP \rightarrow X2\ PP$
$X2 \rightarrow Verb\ NP$
$VP \rightarrow Verb\ PP$
$VP \rightarrow VP\ PP$
$PP \rightarrow Preposition\ NP$

Lexicon

$Det \rightarrow that \mid this \mid the \mid a$
$Noun \rightarrow book \mid flight \mid meal \mid money$
$Verb \rightarrow book \mid include \mid prefer$
$Pronoun \rightarrow I \mid she \mid me$
$Proper\text{-}Noun \rightarrow Houston \mid NWA$
$Aux \rightarrow does$
$Preposition \rightarrow from \mid to \mid on \mid near \mid through$

Exercise

Book the flight through Houston

```
function CKY-PARSE(words, grammar) returns table

  for j ← from 1 to LENGTH(words) do
     for all {A | A → words[j] ∈ grammar}
          table[j − 1, j] ← table[j − 1, j] ∪ A
     for i ← from j − 2 down to 0 do
        for k ← i + 1 to j − 1 do
           for all {A | A → BC ∈ grammar and B ∈ table[i,k] and C ∈ table[k, j]}
                table[i,j] ← table[i,j] ∪ A
```

j=2

words

Book	the	flight	trough	Houston
S, VP, Nominal Noun, Verb [0,1]	[0,2]	[0,3]	[0,4]	[0,5]
	Det [1,2]	[1,3]	[1,4]	[1,5]
		[2,3]	[2,4]	[2,5]
			[3,4]	[3,5]
				[4,5]

$S \rightarrow NP\ VP$
$S \rightarrow X1\ VP$
$X1 \rightarrow Aux\ NP$
$S \rightarrow book \mid include \mid prefer$
$S \rightarrow Verb\ NP$
$S \rightarrow X2\ PP$
$S \rightarrow Verb\ PP$
$S \rightarrow VP\ PP$
$NP \rightarrow I \mid she \mid me$
$NP \rightarrow TWA \mid Houston$
$NP \rightarrow Det\ Nominal$
$Nominal \rightarrow book \mid flight \mid meal \mid money$
$Nominal \rightarrow Nominal\ Noun$
$Nominal \rightarrow Nominal\ PP$
$VP \rightarrow book \mid include \mid prefer$
$VP \rightarrow Verb\ NP$
$VP \rightarrow X2\ PP$
$X2 \rightarrow Verb\ NP$
$VP \rightarrow Verb\ PP$
$VP \rightarrow VP\ PP$
$PP \rightarrow Preposition\ NP$

Lexicon

$Det \rightarrow that \mid this \mid the \mid a$
$Noun \rightarrow book \mid flight \mid meal \mid money$
$Verb \rightarrow book \mid include \mid prefer$
$Pronoun \rightarrow I \mid she \mid me$
$Proper\text{-}Noun \rightarrow Houston \mid NWA$
$Aux \rightarrow does$
$Preposition \rightarrow from \mid to \mid on \mid near \mid through$

Exercise

Book the flight through Houston

```
function CKY-PARSE(words, grammar) returns table

  for j ← from 1 to LENGTH(words) do
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          table[j − 1, j] ← table[j − 1, j] ∪ A
  for i ← from j − 2 down to 0 do
      for k ← i + 1 to j − 1 do
          for all {A | A → BC ∈ grammar and B ∈ table[i,k] and C ∈ table[k, j]}
              table[i,j] ← table[i,j] ∪ A
```

j=2

i=0

words

Book	the	flight	trough	Houston
S, VP, Nominal Noun, Verb [0,1]	[0,2]	[0,3]	[0,4]	[0,5]
	Det [1,2]	[1,3]	[1,4]	[1,5]
		[2,3]	[2,4]	[2,5]
			[3,4]	[3,5]
				[4,5]

$S \rightarrow NP\ VP$
$S \rightarrow X1\ VP$
$X1 \rightarrow Aux\ NP$
$S \rightarrow book \mid include \mid prefer$
$S \rightarrow Verb\ NP$
$S \rightarrow X2\ PP$
$S \rightarrow Verb\ PP$
$S \rightarrow VP\ PP$
$NP \rightarrow I \mid she \mid me$
$NP \rightarrow TWA \mid Houston$
$NP \rightarrow Det\ Nominal$
$Nominal \rightarrow book \mid flight \mid meal \mid money$
$Nominal \rightarrow Nominal\ Noun$
$Nominal \rightarrow Nominal\ PP$
$VP \rightarrow book \mid include \mid prefer$
$VP \rightarrow Verb\ NP$
$VP \rightarrow X2\ PP$
$X2 \rightarrow Verb\ NP$
$VP \rightarrow Verb\ PP$
$VP \rightarrow VP\ PP$
$PP \rightarrow Preposition\ NP$

Lexicon

$Det \rightarrow that \mid this \mid the \mid a$
$Noun \rightarrow book \mid flight \mid meal \mid money$
$Verb \rightarrow book \mid include \mid prefer$
$Pronoun \rightarrow I \mid she \mid me$
$Proper\text{-}Noun \rightarrow Houston \mid NWA$
$Aux \rightarrow does$
$Preposition \rightarrow from \mid to \mid on \mid near \mid through$

Exercise

Book the flight through Houston

```
function CKY-PARSE(words, grammar) returns table

  for j ← from 1 to LENGTH(words) do
     for all {A | A → words[j] ∈ grammar}
           table[j − 1, j] ← table[j − 1, j] ∪ A
     for i ← from j − 2 down to 0 do
        for k ← i + 1 to j − 1 do
           for all {A | A → BC ∈ grammar and B ∈ table[i,k] and C ∈ table[k, j]}
                 table[i,j] ← table[i,j] ∪ A
```

j=2

i=0

k=1

words

Book	*the*	*flight*	*trough*	*Houston*
S, VP, Nominal Noun, Verb [0,1]	[0,2]	[0,3]	[0,4]	[0,5]
	Det [1,2]	[1,3]	[1,4]	[1,5]
		[2,3]	[2,4]	[2,5]
			[3,4]	[3,5]
				[4,5]

no derivations of the form * Det on the right side

$S \rightarrow NP\ VP$
$S \rightarrow X1\ VP$
$X1 \rightarrow Aux\ NP$
$S \rightarrow book \mid include \mid prefer$
$S \rightarrow Verb\ NP$
$S \rightarrow X2\ PP$
$S \rightarrow Verb\ PP$
$S \rightarrow VP\ PP$
$NP \rightarrow I \mid she \mid me$
$NP \rightarrow TWA \mid Houston$
$NP \rightarrow Det\ Nominal$
$Nominal \rightarrow book \mid flight \mid meal \mid money$
$Nominal \rightarrow Nominal\ Noun$
$Nominal \rightarrow Nominal\ PP$
$VP \rightarrow book \mid include \mid prefer$
$VP \rightarrow Verb\ NP$
$VP \rightarrow X2\ PP$
$X2 \rightarrow Verb\ NP$
$VP \rightarrow Verb\ PP$
$VP \rightarrow VP\ PP$
$PP \rightarrow Preposition\ NP$

Lexicon

$Det \rightarrow that \mid this \mid the \mid a$
$Noun \rightarrow book \mid flight \mid meal \mid money$
$Verb \rightarrow book \mid include \mid prefer$
$Pronoun \rightarrow I \mid she \mid me$
$Proper\text{-}Noun \rightarrow Houston \mid NWA$
$Aux \rightarrow does$
$Preposition \rightarrow from \mid to \mid on \mid near \mid through$

Exercise

Book the flight through Houston

```
function CKY-PARSE(words, grammar) returns table

  for j ← from 1 to LENGTH(words) do
      for all {A | A → words[j] ∈ grammar}
            table[j − 1, j] ← table[j − 1, j] ∪ A
      for i ← from j − 2 down to 0 do
          for k ← i + 1 to j − 1 do
              for all {A | A → BC ∈ grammar and B ∈ table[i,k] and C ∈ table[k, j]}
                    table[i,j] ← table[i,j] ∪ A
```

j=3

words

Book	the	flight	trough	Houston
S, VP, Nominal Noun, Verb [0,1]	[0,2]	[0,3]	[0,4]	[0,5]
	Det [1,2]	[1,3]	[1,4]	[1,5]
		Nominal Noun [2,3]	[2,4]	[2,5]
			[3,4]	[3,5]
				[4,5]

$S \rightarrow NP\ VP$
$S \rightarrow X1\ VP$
$X1 \rightarrow Aux\ NP$
$S \rightarrow book \mid include \mid prefer$
$S \rightarrow Verb\ NP$
$S \rightarrow X2\ PP$
$S \rightarrow Verb\ PP$
$S \rightarrow VP\ PP$
$NP \rightarrow I \mid she \mid me$
$NP \rightarrow TWA \mid Houston$
$NP \rightarrow Det\ Nominal$
$Nominal \rightarrow book \mid flight \mid meal \mid money$
$Nominal \rightarrow Nominal\ Noun$
$Nominal \rightarrow Nominal\ PP$
$VP \rightarrow book \mid include \mid prefer$
$VP \rightarrow Verb\ NP$
$VP \rightarrow X2\ PP$
$X2 \rightarrow Verb\ NP$
$VP \rightarrow Verb\ PP$
$VP \rightarrow VP\ PP$
$PP \rightarrow Preposition\ NP$

Lexicon

$Det \rightarrow that \mid this \mid the \mid a$
$Noun \rightarrow book \mid flight \mid meal \mid money$
$Verb \rightarrow book \mid include \mid prefer$
$Pronoun \rightarrow I \mid she \mid me$
$Proper\text{-}Noun \rightarrow Houston \mid NWA$
$Aux \rightarrow does$
$Preposition \rightarrow from \mid to \mid on \mid near \mid through$

Exercise

Book the flight through Houston

```
function CKY-PARSE(words, grammar) returns table

  for j ← from 1 to LENGTH(words) do
      for all {A | A → words[j] ∈ grammar}
          table[j − 1, j] ← table[j − 1, j] ∪ A
  for i ← from j − 2 down to 0 do
      for k ← i + 1 to j − 1 do
          for all {A | A → BC ∈ grammar and B ∈ table[i,k] and C ∈ table[k, j]}
              table[i,j] ← table[i,j] ∪ A
```

j=3

i= 1

words

Book	the	flight	trough	Houston
S, VP, Nominal Noun, Verb [0,1]	[0,2]	[0,3]	[0,4]	[0,5]
	Det [1,2]	[1,3]	[1,4]	[1,5]
		Nominal Noun [2,3]	[2,4]	[2,5]
			[3,4]	[3,5]
				[4,5]

$S \rightarrow NP\ VP$
$S \rightarrow X1\ VP$
$X1 \rightarrow Aux\ NP$
$S \rightarrow book \mid include \mid prefer$
$S \rightarrow Verb\ NP$
$S \rightarrow X2\ PP$
$S \rightarrow Verb\ PP$
$S \rightarrow VP\ PP$
$NP \rightarrow I \mid she \mid me$
$NP \rightarrow TWA \mid Houston$
$NP \rightarrow Det\ Nominal$
$Nominal \rightarrow book \mid flight \mid meal \mid money$
$Nominal \rightarrow Nominal\ Noun$
$Nominal \rightarrow Nominal\ PP$
$VP \rightarrow book \mid include \mid prefer$
$VP \rightarrow Verb\ NP$
$VP \rightarrow X2\ PP$
$X2 \rightarrow Verb\ NP$
$VP \rightarrow Verb\ PP$
$VP \rightarrow VP\ PP$
$PP \rightarrow Preposition\ NP$

Lexicon

$Det \rightarrow that \mid this \mid the \mid a$
$Noun \rightarrow book \mid flight \mid meal \mid money$
$Verb \rightarrow book \mid include \mid prefer$
$Pronoun \rightarrow I \mid she \mid me$
$Proper\text{-}Noun \rightarrow Houston \mid NWA$
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$Preposition \rightarrow from \mid to \mid on \mid near \mid through$

Exercise

Book the flight through Houston

```
function CKY-PARSE(words, grammar) returns table

  for j ← from 1 to LENGTH(words) do
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           table[j − 1, j] ← table[j − 1, j] ∪ A
     for i ← from j − 2 down to 0 do
        for k ← i + 1 to j − 1 do
           for all {A | A → BC ∈ grammar and B ∈ table[i,k] and C ∈ table[k, j]}
                 table[i,j] ← table[i,j] ∪ A
```

j=3

i= 1

k=2

words

Book	the	flight	trough	Houston
S, VP, Nominal Noun, Verb [0,1]	[0,2]	[0,3]	[0,4]	[0,5]
	Det [1,2]	NP [1,3]	[1,4]	[1,5]
		Nominal Noun [2,3]	[2,4]	[2,5]
			[3,4]	[3,5]
				[4,5]

$S \rightarrow NP\ VP$
$S \rightarrow X1\ VP$
$X1 \rightarrow Aux\ NP$
$S \rightarrow book \mid include \mid prefer$
$S \rightarrow Verb\ NP$
$S \rightarrow X2\ PP$
$S \rightarrow Verb\ PP$
$S \rightarrow VP\ PP$
$NP \rightarrow I \mid she \mid me$
$NP \rightarrow TWA \mid Houston$
$NP \rightarrow Det\ Nominal$
$Nominal \rightarrow book \mid flight \mid meal \mid money$
$Nominal \rightarrow Nominal\ Noun$
$Nominal \rightarrow Nominal\ PP$
$VP \rightarrow book \mid include \mid prefer$
$VP \rightarrow Verb\ NP$
$VP \rightarrow X2\ PP$
$X2 \rightarrow Verb\ NP$
$VP \rightarrow Verb\ PP$
$VP \rightarrow VP\ PP$
$PP \rightarrow Preposition\ NP$

Lexicon

$Det \rightarrow that \mid this \mid the \mid a$
$Noun \rightarrow book \mid flight \mid meal \mid money$
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Exercise

Book the flight through Houston

```
function CKY-PARSE(words, grammar) returns table

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            table[j − 1, j] ← table[j − 1, j] ∪ A
      for i ← from j − 2 down to 0 do
          for k ← i + 1 to j − 1 do
              for all {A | A → BC ∈ grammar and B ∈ table[i,k] and C ∈ table[k, j]}
                    table[i,j] ← table[i,j] ∪ A
```

j=3

i= 0

words

Book	the	flight	trough	Houston
S, VP, Nominal Noun, Verb [0,1]	[0,2]	[0,3]	[0,4]	[0,5]
	Det [1,2]	NP [1,3]	[1,4]	[1,5]
		Nominal Noun [2,3]	[2,4]	[2,5]
			[3,4]	[3,5]
				[4,5]

$S \rightarrow NP\ VP$
$S \rightarrow X1\ VP$
$X1 \rightarrow Aux\ NP$
$S \rightarrow book \mid include \mid prefer$
$S \rightarrow Verb\ NP$
$S \rightarrow X2\ PP$
$S \rightarrow Verb\ PP$
$S \rightarrow VP\ PP$
$NP \rightarrow I \mid she \mid me$
$NP \rightarrow TWA \mid Houston$
$NP \rightarrow Det\ Nominal$
$Nominal \rightarrow book \mid flight \mid meal \mid money$
$Nominal \rightarrow Nominal\ Noun$
$Nominal \rightarrow Nominal\ PP$
$VP \rightarrow book \mid include \mid prefer$
$VP \rightarrow Verb\ NP$
$VP \rightarrow X2\ PP$
$X2 \rightarrow Verb\ NP$
$VP \rightarrow Verb\ PP$
$VP \rightarrow VP\ PP$
$PP \rightarrow Preposition\ NP$

Lexicon

$Det \rightarrow that \mid this \mid the \mid a$
$Noun \rightarrow book \mid flight \mid meal \mid money$
$Verb \rightarrow book \mid include \mid prefer$
$Pronoun \rightarrow I \mid she \mid me$
$Proper\text{-}Noun \rightarrow Houston \mid NWA$
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$Preposition \rightarrow from \mid to \mid on \mid near \mid through$

Exercise

Book the flight through Houston

```
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     for i ← from j − 2 down to 0 do
        for k ← i + 1 to j − 1 do
           for all {A | A → BC ∈ grammar and B ∈ table[i,k] and C ∈ table[k, j]}
                table[i,j] ← table[i,j] ∪ A
```

j=3

i= 0

k= 1

words

Book	the	flight	trough	Houston
S,VP, Nominal Noun,Verb [0,1]	[0,2]	S,VP,X2 [0,3]	[0,4]	[0,5]
	Det [1,2]	NP [1,3]	[1,4]	[1,5]
		Nominal Noun [2,3]	[2,4]	[2,5]
			[3,4]	[3,5]
				[4,5]

$S \rightarrow NP\ VP$
$S \rightarrow X1\ VP$
$X1 \rightarrow Aux\ NP$
$S \rightarrow book \mid include \mid prefer$
$S \rightarrow Verb\ NP$
$S \rightarrow X2\ PP$
$S \rightarrow Verb\ PP$
$S \rightarrow VP\ PP$
$NP \rightarrow I \mid she \mid me$
$NP \rightarrow TWA \mid Houston$
$NP \rightarrow Det\ Nominal$
$Nominal \rightarrow book \mid flight \mid meal \mid money$
$Nominal \rightarrow Nominal\ Noun$
$Nominal \rightarrow Nominal\ PP$
$VP \rightarrow book \mid include \mid prefer$
$VP \rightarrow Verb\ NP$
$VP \rightarrow X2\ PP$
$X2 \rightarrow Verb\ NP$
$VP \rightarrow Verb\ PP$
$VP \rightarrow VP\ PP$
$PP \rightarrow Preposition\ NP$

Lexicon

$Det \rightarrow that \mid this \mid the \mid a$
$Noun \rightarrow book \mid flight \mid meal \mid money$
$Verb \rightarrow book \mid include \mid prefer$
$Pronoun \rightarrow I \mid she \mid me$
$Proper\text{-}Noun \rightarrow Houston \mid NWA$
$Aux \rightarrow does$
$Preposition \rightarrow from \mid to \mid on \mid near \mid through$

Exercise

Book the flight through Houston

```
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     for i ← from j − 2 down to 0 do
        for k ← i + 1 to j − 1 do
           for all {A | A → BC ∈ grammar and B ∈ table[i,k] and C ∈ table[k, j]}
                table[i,j] ← table[i,j] ∪ A
```

j=3

i= 0

k= 1

words

Book	the	flight	trough	Houston
S,VP, Nominal Noun,Verb [0,1]	[0,2]	S,VP,X2 [0,3]	[0,4]	[0,5]
	Det [1,2]	NP [1,3]	[1,4]	[1,5]
		Nominal Noun [2,3]	[2,4]	[2,5]
			[3,4]	[3,5]
				[4,5]

$S \rightarrow NP\ VP$
$S \rightarrow X1\ VP$
$X1 \rightarrow Aux\ NP$
$S \rightarrow book \mid include \mid prefer$
$S \rightarrow Verb\ NP$
$S \rightarrow X2\ PP$
$S \rightarrow Verb\ PP$
$S \rightarrow VP\ PP$
$NP \rightarrow I \mid she \mid me$
$NP \rightarrow TWA \mid Houston$
$NP \rightarrow Det\ Nominal$
$Nominal \rightarrow book \mid flight \mid meal \mid money$
$Nominal \rightarrow Nominal\ Noun$
$Nominal \rightarrow Nominal\ PP$
$VP \rightarrow book \mid include \mid prefer$
$VP \rightarrow Verb\ NP$
$VP \rightarrow X2\ PP$
$X2 \rightarrow Verb\ NP$
$VP \rightarrow Verb\ PP$
$VP \rightarrow VP\ PP$
$PP \rightarrow Preposition\ NP$

Lexicon

$Det \rightarrow that \mid this \mid the \mid a$
$Noun \rightarrow book \mid flight \mid meal \mid money$
$Verb \rightarrow book \mid include \mid prefer$
$Pronoun \rightarrow I \mid she \mid me$
$Proper\text{-}Noun \rightarrow Houston \mid NWA$
$Aux \rightarrow does$
$Preposition \rightarrow from \mid to \mid on \mid near \mid through$

Exercise

Book the flight through Houston

```
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     for i ← from j − 2 down to 0 do
        for k ← i + 1 to j − 1 do
           for all {A | A → BC ∈ grammar and B ∈ table[i,k] and C ∈ table[k, j]}
                 table[i,j] ← table[i,j] ∪ A
```

j=3

i= 0

k= 2

words

Book	the	flight	trough	Houston
S,VP, Nominal Noun,Verb [0,1]	[0,2]	S,VP,X2 [0,3]	[0,4]	[0,5]
	Det [1,2]	NP [1,3]	[1,4]	[1,5]
		Nominal Noun [2,3]	[2,4]	[2,5]
			[3,4]	[3,5]
				[4,5]

no symbols in [0,2] we skip

$S \rightarrow NP\ VP$
$S \rightarrow X1\ VP$
$X1 \rightarrow Aux\ NP$
$S \rightarrow book \mid include \mid prefer$
$S \rightarrow Verb\ NP$
$S \rightarrow X2\ PP$
$S \rightarrow Verb\ PP$
$S \rightarrow VP\ PP$
$NP \rightarrow I \mid she \mid me$
$NP \rightarrow TWA \mid Houston$
$NP \rightarrow Det\ Nominal$
$Nominal \rightarrow book \mid flight \mid meal \mid money$
$Nominal \rightarrow Nominal\ Noun$
$Nominal \rightarrow Nominal\ PP$
$VP \rightarrow book \mid include \mid prefer$
$VP \rightarrow Verb\ NP$
$VP \rightarrow X2\ PP$
$X2 \rightarrow Verb\ NP$
$VP \rightarrow Verb\ PP$
$VP \rightarrow VP\ PP$
$PP \rightarrow Preposition\ NP$

Lexicon

$Det \rightarrow that \mid this \mid the \mid a$
$Noun \rightarrow book \mid flight \mid meal \mid money$
$Verb \rightarrow book \mid include \mid prefer$
$Pronoun \rightarrow I \mid she \mid me$
$Proper\text{-}Noun \rightarrow Houston \mid NWA$
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$Preposition \rightarrow from \mid to \mid on \mid near \mid through$

Exercise

Book the flight through Houston

```
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           table[j − 1, j] ← table[j − 1, j] ∪ A
     for i ← from j − 2 down to 0 do
        for k ← i + 1 to j − 1 do
           for all {A | A → BC ∈ grammar and B ∈ table[i,k] and C ∈ table[k, j]}
                 table[i,j] ← table[i,j] ∪ A
```

j=3

i= 0

words

Book	the	flight	trough	Houston
S,VP, Nominal Noun,Verb [0,1]	[0,2]	S,VP,X2 [0,3]	[0,4]	[0,5]
	Det [1,2]	NP [1,3]	[1,4]	[1,5]
		Nominal Noun [2,3]	[2,4]	[2,5]
			[3,4]	[3,5]
				[4,5]

$S \rightarrow NP\ VP$
$S \rightarrow X1\ VP$
$X1 \rightarrow Aux\ NP$
$S \rightarrow book \mid include \mid prefer$
$S \rightarrow Verb\ NP$
$S \rightarrow X2\ PP$
$S \rightarrow Verb\ PP$
$S \rightarrow VP\ PP$
$NP \rightarrow I \mid she \mid me$
$NP \rightarrow TWA \mid Houston$
$NP \rightarrow Det\ Nominal$
$Nominal \rightarrow book \mid flight \mid meal \mid money$
$Nominal \rightarrow Nominal\ Noun$
$Nominal \rightarrow Nominal\ PP$
$VP \rightarrow book \mid include \mid prefer$
$VP \rightarrow Verb\ NP$
$VP \rightarrow X2\ PP$
$X2 \rightarrow Verb\ NP$
$VP \rightarrow Verb\ PP$
$VP \rightarrow VP\ PP$
$PP \rightarrow Preposition\ NP$

Lexicon

$Det \rightarrow that \mid this \mid the \mid a$
$Noun \rightarrow book \mid flight \mid meal \mid money$
$Verb \rightarrow book \mid include \mid prefer$
$Pronoun \rightarrow I \mid she \mid me$
$Proper\text{-}Noun \rightarrow Houston \mid NWA$
$Aux \rightarrow does$
$Preposition \rightarrow from \mid to \mid on \mid near \mid through$

Exercise

Book the flight through Houston

```
function CKY-PARSE(words, grammar) returns table

  for j ← from 1 to LENGTH(words) do
      for all {A | A → words[j] ∈ grammar}
          table[j − 1, j] ← table[j − 1, j] ∪ A
      for i ← from j − 2 down to 0 do
          for k ← i + 1 to j − 1 do
              for all {A | A → BC ∈ grammar and B ∈ table[i,k] and C ∈ table[k, j]}
                  table[i,j] ← table[i,j] ∪ A
```

j=4

words

Book	the	flight	trough	Houston
S,VP, Nominal Noun,Verb [0,1]	[0,2]	S,VP,X2 [0,3]	[0,4]	[0,5]
	Det [1,2]	NP [1,3]	[1,4]	[1,5]
		Nominal Noun [2,3]	[2,4]	[2,5]
			Preposition [3,4]	[3,5]
				[4,5]

$S \rightarrow NP\ VP$
$S \rightarrow X1\ VP$
$X1 \rightarrow Aux\ NP$
$S \rightarrow book \mid include \mid prefer$
$S \rightarrow Verb\ NP$
$S \rightarrow X2\ PP$
$S \rightarrow Verb\ PP$
$S \rightarrow VP\ PP$
$NP \rightarrow I \mid she \mid me$
$NP \rightarrow TWA \mid Houston$
$NP \rightarrow Det\ Nominal$
$Nominal \rightarrow book \mid flight \mid meal \mid money$
$Nominal \rightarrow Nominal\ Noun$
$Nominal \rightarrow Nominal\ PP$
$VP \rightarrow book \mid include \mid prefer$
$VP \rightarrow Verb\ NP$
$VP \rightarrow X2\ PP$
$X2 \rightarrow Verb\ NP$
$VP \rightarrow Verb\ PP$
$VP \rightarrow VP\ PP$
$PP \rightarrow Preposition\ NP$

Lexicon

$Det \rightarrow that \mid this \mid the \mid a$
$Noun \rightarrow book \mid flight \mid meal \mid money$
$Verb \rightarrow book \mid include \mid prefer$
$Pronoun \rightarrow I \mid she \mid me$
$Proper\text{-}Noun \rightarrow Houston \mid NWA$
$Aux \rightarrow does$
$Preposition \rightarrow from \mid to \mid on \mid near \mid through$

Exercise

Book the flight through Houston

```
function CKY-PARSE(words, grammar) returns table

  for j ← from 1 to LENGTH(words) do
      for all {A | A → words[j] ∈ grammar}
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      for k ← i + 1 to j − 1 do
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              table[i,j] ← table[i,j] ∪ A
```

j=4

i= 2

words

Book	the	flight	trough	Houston
S,VP, Nominal Noun,Verb [0,1]	[0,2]	S,VP,X2 [0,3]	[0,4]	[0,5]
	Det [1,2]	NP [1,3]	[1,4]	[1,5]
		Nominal Noun [2,3]	[2,4]	[2,5]
			Preposition [3,4]	[3,5]
				[4,5]

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$S \rightarrow X2\ PP$
$S \rightarrow Verb\ PP$
$S \rightarrow VP\ PP$
$NP \rightarrow I \mid she \mid me$
$NP \rightarrow TWA \mid Houston$
$NP \rightarrow Det\ Nominal$
$Nominal \rightarrow book \mid flight \mid meal \mid money$
$Nominal \rightarrow Nominal\ Noun$
$Nominal \rightarrow Nominal\ PP$
$VP \rightarrow book \mid include \mid prefer$
$VP \rightarrow Verb\ NP$
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$X2 \rightarrow Verb\ NP$
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Book the flight through Houston

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j=4

i= 2

k= 3

words

Book	the	flight	trough	Houston
S,VP, Nominal Noun,Verb [0,1]	[0,2]	S,VP,X2 [0,3]	[0,4]	[0,5]
	Det [1,2]	NP [1,3]	[1,4]	[1,5]
		Nominal Noun [2,3]	[2,4]	[2,5]
			Preposition [3,4]	[3,5]
				[4,5]

no derivations of the form B Preposition on the right side

$S \rightarrow NP\ VP$
$S \rightarrow X1\ VP$
$X1 \rightarrow Aux\ NP$
$S \rightarrow book \mid include \mid prefer$
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$S \rightarrow Verb\ PP$
$S \rightarrow VP\ PP$
$NP \rightarrow I \mid she \mid me$
$NP \rightarrow TWA \mid Houston$
$NP \rightarrow Det\ Nominal$
$Nominal \rightarrow book \mid flight \mid meal \mid money$
$Nominal \rightarrow Nominal\ Noun$
$Nominal \rightarrow Nominal\ PP$
$VP \rightarrow book \mid include \mid prefer$
$VP \rightarrow Verb\ NP$
$VP \rightarrow X2\ PP$
$X2 \rightarrow Verb\ NP$
$VP \rightarrow Verb\ PP$
$VP \rightarrow VP\ PP$
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Exercise

Book the flight through Houston

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           for all {A | A → BC ∈ grammar and B ∈ table[i,k] and C ∈ table[k, j]}
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j=4

i= 1

words

Book	the	flight	trough	Houston
S,VP, Nominal Noun,Verb [0,1]	[0,2]	S,VP,X2 [0,3]	[0,4]	[0,5]
	Det [1,2]	NP [1,3]	[1,4]	[1,5]
		Nominal Noun [2,3]	[2,4]	[2,5]
			Preposition [3,4]	[3,5]
				[4,5]

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$NP \rightarrow Det\ Nominal$
$Nominal \rightarrow book \mid flight \mid meal \mid money$
$Nominal \rightarrow Nominal\ Noun$
$Nominal \rightarrow Nominal\ PP$
$VP \rightarrow book \mid include \mid prefer$
$VP \rightarrow Verb\ NP$
$VP \rightarrow X2\ PP$
$X2 \rightarrow Verb\ NP$
$VP \rightarrow Verb\ PP$
$VP \rightarrow VP\ PP$
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Exercise

Book the flight through Houston

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j=4

i= 1

k= 2

words

Book	the	flight	trough	Houston
S,VP, Nominal Noun,Verb [0,1]	[0,2]	S,VP,X2 [0,3]	[0,4]	[0,5]
	Det [1,2]	NP [1,3]	[1,4]	[1,5]
		Nominal Noun [2,3]	[2,4]	[2,5]
			Preposition [3,4]	[3,5]
				[4,5]

no symbols in [2,4] we skip

$S \rightarrow NP\ VP$
$S \rightarrow X1\ VP$
$X1 \rightarrow Aux\ NP$
$S \rightarrow book \mid include \mid prefer$
$S \rightarrow Verb\ NP$
$S \rightarrow X2\ PP$
$S \rightarrow Verb\ PP$
$S \rightarrow VP\ PP$
$NP \rightarrow I \mid she \mid me$
$NP \rightarrow TWA \mid Houston$
$NP \rightarrow Det\ Nominal$
$Nominal \rightarrow book \mid flight \mid meal \mid money$
$Nominal \rightarrow Nominal\ Noun$
$Nominal \rightarrow Nominal\ PP$
$VP \rightarrow book \mid include \mid prefer$
$VP \rightarrow Verb\ NP$
$VP \rightarrow X2\ PP$
$X2 \rightarrow Verb\ NP$
$VP \rightarrow Verb\ PP$
$VP \rightarrow VP\ PP$
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Exercise

Book the flight through Houston

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j=4

i= 1

k= 3

words

Book	the	flight	trough	Houston
S,VP, Nominal Noun,Verb [0,1]	[0,2]	S,VP,X2 [0,3]	[0,4]	[0,5]
	Det [1,2]	NP [1,3]	[1,4]	[1,5]
		Nominal Noun [2,3]	[2,4]	[2,5]
			Preposition [3,4]	[3,5]
				[4,5]

no derivation of the form * Preposition on the right side

$S \rightarrow NP\ VP$
$S \rightarrow X1\ VP$
$X1 \rightarrow Aux\ NP$
$S \rightarrow book \mid include \mid prefer$
$S \rightarrow Verb\ NP$
$S \rightarrow X2\ PP$
$S \rightarrow Verb\ PP$
$S \rightarrow VP\ PP$
$NP \rightarrow I \mid she \mid me$
$NP \rightarrow TWA \mid Houston$
$NP \rightarrow Det\ Nominal$
$Nominal \rightarrow book \mid flight \mid meal \mid money$
$Nominal \rightarrow Nominal\ Noun$
$Nominal \rightarrow Nominal\ PP$
$VP \rightarrow book \mid include \mid prefer$
$VP \rightarrow Verb\ NP$
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$X2 \rightarrow Verb\ NP$
$VP \rightarrow Verb\ PP$
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Lexicon

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Exercise

Book the flight through Houston

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          for all {A | A → BC ∈ grammar and B ∈ table[i,k] and C ∈ table[k, j]}
              table[i,j] ← table[i,j] ∪ A
```

j=4

i= 0

words

Book	the	flight	trough	Houston
S,VP, Nominal Noun,Verb [0,1]	[0,2]	S,VP,X2 [0,3]	[0,4]	[0,5]
	Det [1,2]	NP [1,3]	[1,4]	[1,5]
		Nominal Noun [2,3]	[2,4]	[2,5]
			Preposition [3,4]	[3,5]
				[4,5]

$S \rightarrow NP\ VP$
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$Nominal \rightarrow Nominal\ Noun$
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Book the flight through Houston

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j=4

i= 0

k= 1

words

Book	the	flight	trough	Houston
S,VP, Nominal Noun,Verb [0,1]	[0,2]	S,VP,X2 [0,3]	[0,4]	[0,5]
	Det [1,2]	NP [1,3]	[1,4]	[1,5]
		Nominal Noun [2,3]	[2,4]	[2,5]
			Preposition [3,4]	[3,5]
				[4,5]

no symbols in [1,4] we skip

$S \rightarrow NP\ VP$
$S \rightarrow X1\ VP$
$X1 \rightarrow Aux\ NP$
$S \rightarrow book \mid include \mid prefer$
$S \rightarrow Verb\ NP$
$S \rightarrow X2\ PP$
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$NP \rightarrow I \mid she \mid me$
$NP \rightarrow TWA \mid Houston$
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$Nominal \rightarrow Nominal\ Noun$
$Nominal \rightarrow Nominal\ PP$
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Book the flight through Houston

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```

j=4

i= 0

k= 2

words

Book	the	flight	trough	Houston
S,VP, Nominal Noun,Verb [0,1]	[0,2]	S,VP,X2 [0,3]	[0,4]	[0,5]
	Det [1,2]	NP [1,3]	[1,4]	[1,5]
		Nominal Noun [2,3]	[2,4]	[2,5]
			Preposition [3,4]	[3,5]
				[4,5]

no symbols in [1,2] and in [2,4] we skip

$S \rightarrow NP\ VP$
$S \rightarrow X1\ VP$
$X1 \rightarrow Aux\ NP$
$S \rightarrow book \mid include \mid prefer$
$S \rightarrow Verb\ NP$
$S \rightarrow X2\ PP$
$S \rightarrow Verb\ PP$
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$Nominal \rightarrow Nominal\ Noun$
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$X2 \rightarrow Verb\ NP$
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j=4

i= 0

k= 3

words

Book	the	flight	trough	Houston
S,VP, Nominal Noun,Verb [0,1]	[0,2]	S,VP,X2 [0,3]	[0,4]	[0,5]
	Det [1,2]	NP [1,3]	[1,4]	[1,5]
		Nominal Noun [2,3]	[2,4]	[2,5]
			Preposition [3,4]	[3,5]
				[4,5]

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$S \rightarrow Verb\ NP$
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$S \rightarrow Verb\ PP$
$S \rightarrow VP\ PP$
$NP \rightarrow I \mid she \mid me$
$NP \rightarrow TWA \mid Houston$
$NP \rightarrow Det\ Nominal$
$Nominal \rightarrow book \mid flight \mid meal \mid money$
$Nominal \rightarrow Nominal\ Noun$
$Nominal \rightarrow Nominal\ PP$
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$X2 \rightarrow Verb\ NP$
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j=4

i= 0

words

Book	the	flight	trough	Houston
S,VP, Nominal Noun,Verb [0,1]	[0,2]	S,VP,X2 [0,3]	[0,4]	[0,5]
	Det [1,2]	NP [1,3]	[1,4]	[1,5]
		Nominal Noun [2,3]	[2,4]	[2,5]
			Preposition [3,4]	[3,5]
				[4,5]

$S \rightarrow NP\ VP$
$S \rightarrow X1\ VP$
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$S \rightarrow VP\ PP$
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$Nominal \rightarrow Nominal\ PP$
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$VP \rightarrow Verb\ NP$
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$X2 \rightarrow Verb\ NP$
$VP \rightarrow Verb\ PP$
$VP \rightarrow VP\ PP$
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Lexicon

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Book the flight through Houston

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              for all {A | A → BC ∈ grammar and B ∈ table[i,k] and C ∈ table[k, j]}
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```

j=5

words

Book	the	flight	trough	Houston
S,VP, Nominal Noun,Verb [0,1]	[0,2]	S,VP,X2 [0,3]	[0,4]	[0,5]
	Det [1,2]	NP [1,3]	[1,4]	[1,5]
		Nominal Noun [2,3]	[2,4]	[2,5]
			Preposition [3,4]	[3,5]
				NP, Proper-Noun [4,5]

$S \rightarrow NP\ VP$
$S \rightarrow X1\ VP$
$X1 \rightarrow Aux\ NP$
$S \rightarrow book \mid include \mid prefer$
$S \rightarrow Verb\ NP$
$S \rightarrow X2\ PP$
$S \rightarrow Verb\ PP$
$S \rightarrow VP\ PP$
$NP \rightarrow I \mid she \mid me$
$NP \rightarrow TWA \mid Houston$
$NP \rightarrow Det\ Nominal$
$Nominal \rightarrow book \mid flight \mid meal \mid money$
$Nominal \rightarrow Nominal\ Noun$
$Nominal \rightarrow Nominal\ PP$
$VP \rightarrow book \mid include \mid prefer$
$VP \rightarrow Verb\ NP$
$VP \rightarrow X2\ PP$
$X2 \rightarrow Verb\ NP$
$VP \rightarrow Verb\ PP$
$VP \rightarrow VP\ PP$
$PP \rightarrow Preposition\ NP$

Lexicon

$Det \rightarrow that \mid this \mid the \mid a$
$Noun \rightarrow book \mid flight \mid meal \mid money$
$Verb \rightarrow book \mid include \mid prefer$
$Pronoun \rightarrow I \mid she \mid me$
$Proper\text{-}Noun \rightarrow Houston \mid NWA$
$Aux \rightarrow does$
$Preposition \rightarrow from \mid to \mid on \mid near \mid through$

Exercise

Book the flight through Houston

```
function CKY-PARSE(words, grammar) returns table

  for j ← from 1 to LENGTH(words) do
      for all {A | A → words[j] ∈ grammar}
          table[j − 1, j] ← table[j − 1, j] ∪ A
      for i ← from j − 2 down to 0 do
          for k ← i + 1 to j − 1 do
              for all {A | A → BC ∈ grammar and B ∈ table[i,k] and C ∈ table[k, j]}
                  table[i,j] ← table[i,j] ∪ A
```

j=5

words

Book	the	flight	trough	Houston
S,VP, Nominal Noun,Verb [0,1]	[0,2]	S,VP,X2 [0,3]	[0,4]	[0,5]
	Det [1,2]	NP [1,3]	[1,4]	[1,5]
		Nominal Noun [2,3]	[2,4]	[2,5]
			Preposition [3,4]	[3,5]
				NP, Proper-Noun [4,5]

$S \rightarrow NP\ VP$
$S \rightarrow X1\ VP$
$X1 \rightarrow Aux\ NP$
$S \rightarrow book \mid include \mid prefer$
$S \rightarrow Verb\ NP$
$S \rightarrow X2\ PP$
$S \rightarrow Verb\ PP$
$S \rightarrow VP\ PP$
$NP \rightarrow I \mid she \mid me$
$NP \rightarrow TWA \mid Houston$
$NP \rightarrow Det\ Nominal$
$Nominal \rightarrow book \mid flight \mid meal \mid money$
$Nominal \rightarrow Nominal\ Noun$
$Nominal \rightarrow Nominal\ PP$
$VP \rightarrow book \mid include \mid prefer$
$VP \rightarrow Verb\ NP$
$VP \rightarrow X2\ PP$
$X2 \rightarrow Verb\ NP$
$VP \rightarrow Verb\ PP$
$VP \rightarrow VP\ PP$
$PP \rightarrow Preposition\ NP$

Lexicon

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Book the flight through Houston

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        for k ← i + 1 to j − 1 do
           for all {A | A → BC ∈ grammar and B ∈ table[i,k] and C ∈ table[k, j]}
                 table[i,j] ← table[i,j] ∪ A
```

j=5

i=3

words

Book	the	flight	trough	Houston
S,VP, Nominal Noun,Verb [0,1]	[0,2]	S,VP,X2 [0,3]	[0,4]	[0,5]
	Det [1,2]	NP [1,3]	[1,4]	[1,5]
		Nominal Noun [2,3]	[2,4]	[2,5]
			Preposition [3,4]	[3,5]
				NP, Proper-Noun [4,5]

$S \rightarrow NP\ VP$
$S \rightarrow X1\ VP$
$X1 \rightarrow Aux\ NP$
$S \rightarrow book \mid include \mid prefer$
$S \rightarrow Verb\ NP$
$S \rightarrow X2\ PP$
$S \rightarrow Verb\ PP$
$S \rightarrow VP\ PP$
$NP \rightarrow I \mid she \mid me$
$NP \rightarrow TWA \mid Houston$
$NP \rightarrow Det\ Nominal$
$Nominal \rightarrow book \mid flight \mid meal \mid money$
$Nominal \rightarrow Nominal\ Noun$
$Nominal \rightarrow Nominal\ PP$
$VP \rightarrow book \mid include \mid prefer$
$VP \rightarrow Verb\ NP$
$VP \rightarrow X2\ PP$
$X2 \rightarrow Verb\ NP$
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Book the flight through Houston

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```

j=5

i=3

k=4

words

Book	the	flight	trough	Houston
S,VP, Nominal Noun,Verb [0,1]	[0,2]	S,VP,X2 [0,3]	[0,4]	[0,5]
	Det [1,2]	NP [1,3]	[1,4]	[1,5]
		Nominal Noun [2,3]	[2,4]	[2,5]
			Preposition [3,4]	PP [3,5]
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$S \rightarrow NP\ VP$
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$S \rightarrow Verb\ NP$
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$S \rightarrow Verb\ PP$
$S \rightarrow VP\ PP$
$NP \rightarrow I \mid she \mid me$
$NP \rightarrow TWA \mid Houston$
$NP \rightarrow Det\ Nominal$
$Nominal \rightarrow book \mid flight \mid meal \mid money$
$Nominal \rightarrow Nominal\ Noun$
$Nominal \rightarrow Nominal\ PP$
$VP \rightarrow book \mid include \mid prefer$
$VP \rightarrow Verb\ NP$
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$VP \rightarrow Verb\ PP$
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Exercise

Book the flight through Houston

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```

j=5

i=2

words

Book	the	flight	trough	Houston
S,VP, Nominal Noun,Verb [0,1]	[0,2]	S,VP,X2 [0,3]	[0,4]	[0,5]
	Det [1,2]	NP [1,3]	[1,4]	[1,5]
		Nominal Noun [2,3]	[2,4]	[2,5]
			Preposition [3,4]	PP [3,5]
				NP, Proper-Noun [4,5]

$S \rightarrow NP\ VP$
$S \rightarrow X1\ VP$
$X1 \rightarrow Aux\ NP$
$S \rightarrow book \mid include \mid prefer$
$S \rightarrow Verb\ NP$
$S \rightarrow X2\ PP$
$S \rightarrow Verb\ PP$
$S \rightarrow VP\ PP$
$NP \rightarrow I \mid she \mid me$
$NP \rightarrow TWA \mid Houston$
$NP \rightarrow Det\ Nominal$
$Nominal \rightarrow book \mid flight \mid meal \mid money$
$Nominal \rightarrow Nominal\ Noun$
$Nominal \rightarrow Nominal\ PP$
$VP \rightarrow book \mid include \mid prefer$
$VP \rightarrow Verb\ NP$
$VP \rightarrow X2\ PP$
$X2 \rightarrow Verb\ NP$
$VP \rightarrow Verb\ PP$
$VP \rightarrow VP\ PP$
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Lexicon

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Exercise

Book the flight through Houston

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                  table[i,j] ← table[i,j] ∪ A
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j=5

i=2

k=3

words

Book	the	flight	trough	Houston
S,VP, Nominal Noun,Verb [0,1]	[0,2]	S,VP,X2 [0,3]	[0,4]	[0,5]
	Det [1,2]	NP [1,3]	[1,4]	[1,5]
		Nominal Noun [2,3]	[2,4]	Nominal [2,5]
			Preposition [3,4]	PP [3,5]
				NP, Proper-Noun [4,5]

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$S \rightarrow X1\ VP$
$X1 \rightarrow Aux\ NP$
$S \rightarrow book \mid include \mid prefer$
$S \rightarrow Verb\ NP$
$S \rightarrow X2\ PP$
$S \rightarrow Verb\ PP$
$S \rightarrow VP\ PP$
$NP \rightarrow I \mid she \mid me$
$NP \rightarrow TWA \mid Houston$
$NP \rightarrow Det\ Nominal$
$Nominal \rightarrow book \mid flight \mid meal \mid money$
$Nominal \rightarrow Nominal\ Noun$
$Nominal \rightarrow Nominal\ PP$
$VP \rightarrow book \mid include \mid prefer$
$VP \rightarrow Verb\ NP$
$VP \rightarrow X2\ PP$
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                  table[i,j] ← table[i,j] ∪ A
```

j=5

i=2

k=4

words

Book	the	flight	trough	Houston
S,VP, Nominal Noun,Verb [0,1]	[0,2]	S,VP,X2 [0,3]	[0,4]	[0,5]
	Det [1,2]	NP [1,3]	[1,4]	[1,5]
		Nominal Noun [2,3]	[2,4]	Nominal [2,5]
			Preposition [3,4]	PP [3,5]
				NP, Proper-Noun [4,5]

no symbols in [2,4] we skip

$S \rightarrow NP\ VP$
$S \rightarrow X1\ VP$
$X1 \rightarrow Aux\ NP$
$S \rightarrow book \mid include \mid prefer$
$S \rightarrow Verb\ NP$
$S \rightarrow X2\ PP$
$S \rightarrow Verb\ PP$
$S \rightarrow VP\ PP$
$NP \rightarrow I \mid she \mid me$
$NP \rightarrow TWA \mid Houston$
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$Nominal \rightarrow book \mid flight \mid meal \mid money$
$Nominal \rightarrow Nominal\ Noun$
$Nominal \rightarrow Nominal\ PP$
$VP \rightarrow book \mid include \mid prefer$
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              for all {A | A → BC ∈ grammar and B ∈ table[i,k] and C ∈ table[k, j]}
                    table[i,j] ← table[i,j] ∪ A
```

j=5

i=1

words

Book	the	flight	trough	Houston
S,VP, Nominal Noun,Verb [0,1]	[0,2]	S,VP,X2 [0,3]	[0,4]	[0,5]
	Det [1,2]	NP [1,3]	[1,4]	[1,5]
		Nominal Noun [2,3]	[2,4]	Nominal [2,5]
			Preposition [3,4]	PP [3,5]
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$Nominal \rightarrow Nominal\ Noun$
$Nominal \rightarrow Nominal\ PP$
$VP \rightarrow book \mid include \mid prefer$
$VP \rightarrow Verb\ NP$
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$X2 \rightarrow Verb\ NP$
$VP \rightarrow Verb\ PP$
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j=5

i=1

k=2

words

Book	the	flight	trough	Houston
S,VP, Nominal Noun,Verb [0,1]	[0,2]	S,VP,X2 [0,3]	[0,4]	[0,5]
	Det [1,2]	NP [1,3]	[1,4]	NP [1,5]
		Nominal Noun [2,3]	[2,4]	Nominal [2,5]
			Preposition [3,4]	PP [3,5]
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$Nominal \rightarrow Nominal\ Noun$
$Nominal \rightarrow Nominal\ PP$
$VP \rightarrow book \mid include \mid prefer$
$VP \rightarrow Verb\ NP$
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j=5

i=1

k=3

words

Book	the	flight	trough	Houston
S,VP, Nominal Noun,Verb [0,1]	[0,2]	S,VP,X2 [0,3]	[0,4]	[0,5]
	Det [1,2]	NP [1,3]	[1,4]	NP [1,5]
		Nominal Noun [2,3]	[2,4]	Nominal [2,5]
			Preposition [3,4]	PP [3,5]
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$S \rightarrow NP\ VP$
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$Nominal \rightarrow book \mid flight \mid meal \mid money$
$Nominal \rightarrow Nominal\ Noun$
$Nominal \rightarrow Nominal\ PP$
$VP \rightarrow book \mid include \mid prefer$
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j=5

i=1

k=4

words

Book	the	flight	trough	Houston
S,VP, Nominal Noun,Verb [0,1]	[0,2]	S,VP,X2 [0,3]	[0,4]	[0,5]
	Det [1,2]	NP [1,3]	[1,4]	NP [1,5]
		Nominal Noun [2,3]	[2,4]	Nominal [2,5]
			Preposition [3,4]	PP [3,5]
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$Nominal \rightarrow Nominal\ Noun$
$Nominal \rightarrow Nominal\ PP$
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$VP \rightarrow Verb\ NP$
$VP \rightarrow X2\ PP$
$X2 \rightarrow Verb\ NP$
$VP \rightarrow Verb\ PP$
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                 table[i,j] ← table[i,j] ∪ A
```

j=5

i=0

words

Book	*the*	*flight*	*trough*	*Houston*
S,VP, Nominal Noun,Verb [0,1]	[0,2]	S,VP,X2 [0,3]	[0,4]	[0,5]
	Det [1,2]	NP [1,3]	[1,4]	NP [1,5]
		Nominal Noun [2,3]	[2,4]	Nominal [2,5]
			Preposition [3,4]	PP [3,5]
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$S \rightarrow NP\ VP$
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$S \rightarrow Verb\ PP$
$S \rightarrow VP\ PP$
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$Nominal \rightarrow book \mid flight \mid meal \mid money$
$Nominal \rightarrow Nominal\ Noun$
$Nominal \rightarrow Nominal\ PP$
$VP \rightarrow book \mid include \mid prefer$
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$X2 \rightarrow Verb\ NP$
$VP \rightarrow Verb\ PP$
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Lexicon

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                 table[i,j] ← table[i,j] ∪ A
```

j=5

i=0

k=1

words

Book	the	flight	trough	Houston
S,VP, Nominal Noun,Verb [0,1]	[0,2]	S,VP,X2 [0,3]	[0,4]	S1,VP, X2 [0,5]
	Det [1,2]	NP [1,3]	[1,4]	NP [1,5]
		Nominal Noun [2,3]	[2,4]	Nominal [2,5]
			Preposition [3,4]	PP [3,5]
				NP, Proper-Noun [4,5]

$S \rightarrow NP\ VP$
$S \rightarrow X1\ VP$
$X1 \rightarrow Aux\ NP$
$S \rightarrow book \mid include \mid prefer$
$S \rightarrow Verb\ NP$
$S \rightarrow X2\ PP$
$S \rightarrow Verb\ PP$
$S \rightarrow VP\ PP$
$NP \rightarrow I \mid she \mid me$
$NP \rightarrow TWA \mid Houston$
$NP \rightarrow Det\ Nominal$
$Nominal \rightarrow book \mid flight \mid meal \mid money$
$Nominal \rightarrow Nominal\ Noun$
$Nominal \rightarrow Nominal\ PP$
$VP \rightarrow book \mid include \mid prefer$
$VP \rightarrow Verb\ NP$
$VP \rightarrow X2\ PP$
$X2 \rightarrow Verb\ NP$
$VP \rightarrow Verb\ PP$
$VP \rightarrow VP\ PP$
$PP \rightarrow Preposition\ NP$

Lexicon

$Det \rightarrow that \mid this \mid the \mid a$
$Noun \rightarrow book \mid flight \mid meal \mid money$
$Verb \rightarrow book \mid include \mid prefer$
$Pronoun \rightarrow I \mid she \mid me$
$Proper\text{-}Noun \rightarrow Houston \mid NWA$
$Aux \rightarrow does$
$Preposition \rightarrow from \mid to \mid on \mid near \mid through$

Exercise

Book the flight through Houston

```
function CKY-PARSE(words, grammar) returns table

  for j ← from 1 to LENGTH(words) do
      for all {A | A → words[j] ∈ grammar}
          table[j − 1, j] ← table[j − 1, j] ∪ A
  for i ← from j − 2 down to 0 do
      for k ← i + 1 to j − 1 do
          for all {A | A → BC ∈ grammar and B ∈ table[i,k] and C ∈ table[k, j]}
              table[i,j] ← table[i,j] ∪ A
```

j=5

i=0

k=2

words

Book	the	flight	trough	Houston
S,VP, Nominal Noun,Verb [0,1]	[0,2]	S,VP,X2 [0,3]	[0,4]	S1,VP, X2 [0,5]
	Det [1,2]	NP [1,3]	[1,4]	NP [1,5]
		Nominal Noun [2,3]	[2,4]	Nominal [2,5]
			Preposition [3,4]	PP [3,5]
				NP, Proper-Noun [4,5]

$S \rightarrow NP\ VP$
$S \rightarrow X1\ VP$
$X1 \rightarrow Aux\ NP$
$S \rightarrow book \mid include \mid prefer$
$S \rightarrow Verb\ NP$
$S \rightarrow X2\ PP$
$S \rightarrow Verb\ PP$
$S \rightarrow VP\ PP$
$NP \rightarrow I \mid she \mid me$
$NP \rightarrow TWA \mid Houston$
$NP \rightarrow Det\ Nominal$
$Nominal \rightarrow book \mid flight \mid meal \mid money$
$Nominal \rightarrow Nominal\ Noun$
$Nominal \rightarrow Nominal\ PP$
$VP \rightarrow book \mid include \mid prefer$
$VP \rightarrow Verb\ NP$
$VP \rightarrow X2\ PP$
$X2 \rightarrow Verb\ NP$
$VP \rightarrow Verb\ PP$
$VP \rightarrow VP\ PP$
$PP \rightarrow Preposition\ NP$

Lexicon

$Det \rightarrow that \mid this \mid the \mid a$
$Noun \rightarrow book \mid flight \mid meal \mid money$
$Verb \rightarrow book \mid include \mid prefer$
$Pronoun \rightarrow I \mid she \mid me$
$Proper\text{-}Noun \rightarrow Houston \mid NWA$
$Aux \rightarrow does$
$Preposition \rightarrow from \mid to \mid on \mid near \mid through$

Exercise

Book the flight through Houston

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function CKY-PARSE(words, grammar) returns table

  for j ← from 1 to LENGTH(words) do
     for all {A | A → words[j] ∈ grammar}
           table[j − 1, j] ← table[j − 1, j] ∪ A
     for i ← from j − 2 down to 0 do
        for k ← i + 1 to j − 1 do
           for all {A | A → BC ∈ grammar and B ∈ table[i,k] and C ∈ table[k, j]}
                 table[i,j] ← table[i,j] ∪ A
```

j=5

i=0

k=3

words

Book	the	flight	trough	Houston
S,VP, Nominal Noun,Verb [0,1]	[0,2]	S,VP,X2 [0,3]	[0,4]	S1,VP, X2 S2,VP S3 [0,5]
	Det [1,2]	NP [1,3]	[1,4]	NP [1,5]
		Nominal Noun [2,3]	[2,4]	Nominal [2,5]
			Preposition [3,4]	PP [3,5]
				NP, Proper-Noun [4,5]

$S \rightarrow NP\ VP$
$S \rightarrow X1\ VP$
$X1 \rightarrow Aux\ NP$
$S \rightarrow book \mid include \mid prefer$
$S \rightarrow Verb\ NP$
$S \rightarrow X2\ PP$
$S \rightarrow Verb\ PP$
$S \rightarrow VP\ PP$
$NP \rightarrow I \mid she \mid me$
$NP \rightarrow TWA \mid Houston$
$NP \rightarrow Det\ Nominal$
$Nominal \rightarrow book \mid flight \mid meal \mid money$
$Nominal \rightarrow Nominal\ Noun$
$Nominal \rightarrow Nominal\ PP$
$VP \rightarrow book \mid include \mid prefer$
$VP \rightarrow Verb\ NP$
$VP \rightarrow X2\ PP$
$X2 \rightarrow Verb\ NP$
$VP \rightarrow Verb\ PP$
$VP \rightarrow VP\ PP$
$PP \rightarrow Preposition\ NP$

Lexicon

$Det \rightarrow that \mid this \mid the \mid a$
$Noun \rightarrow book \mid flight \mid meal \mid money$
$Verb \rightarrow book \mid include \mid prefer$
$Pronoun \rightarrow I \mid she \mid me$
$Proper\text{-}Noun \rightarrow Houston \mid NWA$
$Aux \rightarrow does$
$Preposition \rightarrow from \mid to \mid on \mid near \mid through$

Exercise

Book the flight through Houston

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function CKY-PARSE(words, grammar) returns table

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        for k ← i + 1 to j − 1 do
           for all {A | A → BC ∈ grammar and B ∈ table[i,k] and C ∈ table[k, j]}
                 table[i,j] ← table[i,j] ∪ A
```

j=5

i=0

k=4

words

Book	the	flight	trough	Houston
S,VP, Nominal Noun,Verb [0,1]	[0,2]	S,VP,X2 [0,3]	[0,4]	S1,VP, X2 S2,VP S3 [0,5]
	Det [1,2]	NP [1,3]	[1,4]	NP [1,5]
		Nominal Noun [2,3]	[2,4]	Nominal [2,5]
			Preposition [3,4]	PP [3,5]
				NP, Proper-Noun [4,5]

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$S \rightarrow X1\ VP$
$X1 \rightarrow Aux\ NP$
$S \rightarrow book \mid include \mid prefer$
$S \rightarrow Verb\ NP$
$S \rightarrow X2\ PP$
$S \rightarrow Verb\ PP$
$S \rightarrow VP\ PP$
$NP \rightarrow I \mid she \mid me$
$NP \rightarrow TWA \mid Houston$
$NP \rightarrow Det\ Nominal$
$Nominal \rightarrow book \mid flight \mid meal \mid money$
$Nominal \rightarrow Nominal\ Noun$
$Nominal \rightarrow Nominal\ PP$
$VP \rightarrow book \mid include \mid prefer$
$VP \rightarrow Verb\ NP$
$VP \rightarrow X2\ PP$
$X2 \rightarrow Verb\ NP$
$VP \rightarrow Verb\ PP$
$VP \rightarrow VP\ PP$
$PP \rightarrow Preposition\ NP$

Lexicon

$Det \rightarrow that \mid this \mid the \mid a$
$Noun \rightarrow book \mid flight \mid meal \mid money$
$Verb \rightarrow book \mid include \mid prefer$
$Pronoun \rightarrow I \mid she \mid me$
$Proper\text{-}Noun \rightarrow Houston \mid NWA$
$Aux \rightarrow does$
$Preposition \rightarrow from \mid to \mid on \mid near \mid through$

Exercise

Book the flight through Houston

```
function CKY-PARSE(words, grammar) returns table

  for j ← from 1 to LENGTH(words) do
     for all {A | A → words[j] ∈ grammar}
           table[j − 1, j] ← table[j − 1, j] ∪ A
     for i ← from j − 2 down to 0 do
        for k ← i + 1 to j − 1 do
           for all {A | A → BC ∈ grammar and B ∈ table[i,k] and C ∈ table[k, j]}
                 table[i,j] ← table[i,j] ∪ A
```

words

Book	the	flight	trough	Houston
S,VP, Nominal Noun,Verb [0,1]	[0,2]	S,VP,X2 [0,3]	[0,4]	S1,VP, X2 S2,VP S3 [0,5]
	Det [1,2]	NP [1,3]	[1,4]	NP [1,5]
		Nominal Noun [2,3]	[2,4]	Nominal [2,5]
			Preposition [3,4]	PP [3,5]
				NP, Proper-Noun [4,5]

$S \rightarrow NP\ VP$
$S \rightarrow X1\ VP$
$X1 \rightarrow Aux\ NP$
$S \rightarrow book \mid include \mid prefer$
$S \rightarrow Verb\ NP$
$S \rightarrow X2\ PP$
$S \rightarrow Verb\ PP$
$S \rightarrow VP\ PP$
$NP \rightarrow I \mid she \mid me$
$NP \rightarrow TWA \mid Houston$
$NP \rightarrow Det\ Nominal$
$Nominal \rightarrow book \mid flight \mid meal \mid money$
$Nominal \rightarrow Nominal\ Noun$
$Nominal \rightarrow Nominal\ PP$
$VP \rightarrow book \mid include \mid prefer$
$VP \rightarrow Verb\ NP$
$VP \rightarrow X2\ PP$
$X2 \rightarrow Verb\ NP$
$VP \rightarrow Verb\ PP$
$VP \rightarrow VP\ PP$
$PP \rightarrow Preposition\ NP$

Lexicon

$Det \rightarrow that \mid this \mid the \mid a$
$Noun \rightarrow book \mid flight \mid meal \mid money$
$Verb \rightarrow book \mid include \mid prefer$
$Pronoun \rightarrow I \mid she \mid me$
$Proper\text{-}Noun \rightarrow Houston \mid TWA$
$Aux \rightarrow does$
$Preposition \rightarrow from \mid to \mid on \mid near \mid through$

RECOGNIZED since [0,5] contains ***S***

Exercise

starting from **S1**

s1

words	*Book*	*the*	*flight*	*trough*	*Houston*
	S,VP, Nominal Noun,Verb [0,1]	[0,2]	S,VP,X2 [0,3]	[0,4]	**S1**,VP, X2 S2,VP S3 [0,5]
		Det [1,2]	NP [1,3]	[1,4]	NP [1,5]
			Nominal Noun [2,3]	[2,4]	Nominal [2,5]
				Preposition [3,4]	PP [3,5]
					NP, Proper-Noun [4,5]

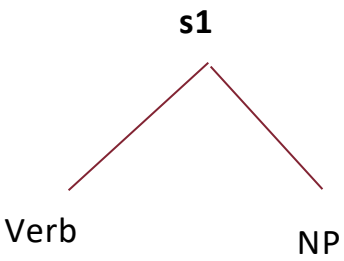
Lexicon
Det → *that* | *this* | *the* | *a*
Noun → *book* | *flight* | *meal* | *money*
Verb → *book* | *include* | *prefer*
Pronoun → *I* | *she* | *me*
Proper-Noun → *Houston* | *NWA*
Aux → *does*
Preposition → *from* | *to* | *on* | *near* | *through*

S → *NP VP*
S → *X1 VP*
X1 → *Aux NP*
S → *book* | *include* | *prefer*
S → *Verb NP*
S → *X2 PP*
S → *Verb PP*
S → *VP PP*
NP → *I* | *she* | *me*
NP → *TWA* | *Houston*
NP → *Det Nominal*
Nominal → *book* | *flight* | *meal* | *money*
Nominal → *Nominal Noun*
Nominal → *Nominal PP*
VP → *book* | *include* | *prefer*
VP → *Verb NP*
VP → *X2 PP*
X2 → *Verb NP*
VP → *Verb PP*
VP → *VP PP*
PP → *Preposition NP*

Exercise

starting from **S1**

words	*Book*	*the*	*flight*	*trough*	*Houston*
	S,VP, Nominal Noun,Verb [0,1]	[0,2]	S,VP,X2 [0,3]	[0,4]	S1,VP, X2 S2,VP S3 [0,5]
		Det [1,2]	NP [1,3]	[1,4]	NP [1,5]
			Nominal Noun [2,3]	[2,4]	Nominal [2,5]
				Preposition [3,4]	PP [3,5]
					NP, Proper-Noun [4,5]

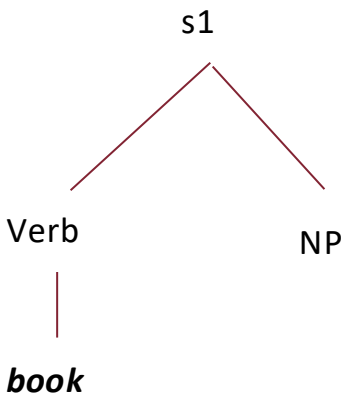


Lexicon
Det → *that* | *this* | *the* | *a*
Noun → *book* | *flight* | *meal* | *money*
Verb → *book* | *include* | *prefer*
Pronoun → *I* | *she* | *me*
Proper-Noun → *Houston* | *NWA*
Aux → *does*
Preposition → *from* | *to* | *on* | *near* | *through*

S → *NP VP*
S → *X1 VP*
X1 → *Aux NP*
S → *book* | *include* | *prefer*
S → *Verb NP*
S → *X2 PP*
S → *Verb PP*
S → *VP PP*
NP → *I* | *she* | *me*
NP → *TWA* | *Houston*
NP → *Det Nominal*
Nominal → *book* | *flight* | *meal* | *money*
Nominal → *Nominal Noun*
Nominal → *Nominal PP*
VP → *book* | *include* | *prefer*
VP → *Verb NP*
VP → *X2 PP*
X2 → *Verb NP*
VP → *Verb PP*
VP → *VP PP*
PP → *Preposition NP*

Exercise

words	***Book***	*the*	*flight*	*trough*	*Houston*
	S,VP, Nominal Noun,**Verb** [0,1]	[0,2]	S,VP,X2 [0,3]	[0,4]	S1,VP, X2 S2,VP S3 [0,5]
		Det [1,2]	NP [1,3]	[1,4]	NP [1,5]
			Nominal Noun [2,3]	[2,4]	Nominal [2,5]
				Preposition [3,4]	PP [3,5]
					NP, Proper-Noun [4,5]

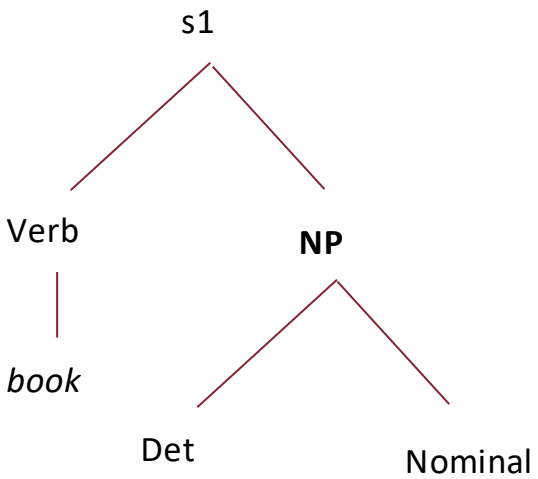


Lexicon
Det → *that* | *this* | *the* | *a*
Noun → *book* | *flight* | *meal* | *money*
Verb → *book* | *include* | *prefer*
Pronoun → *I* | *she* | *me*
Proper-Noun → *Houston* | *NWA*
Aux → *does*
Preposition → *from* | *to* | *on* | *near* | *through*

S → *NP VP*
S → *X1 VP*
X1 → *Aux NP*
S → *book* | *include* | *prefer*
S → *Verb NP*
S → *X2 PP*
S → *Verb PP*
S → *VP PP*
NP → *I* | *she* | *me*
NP → *TWA* | *Houston*
NP → *Det Nominal*
Nominal → *book* | *flight* | *meal* | *money*
Nominal → *Nominal Noun*
Nominal → *Nominal PP*
VP → *book* | *include* | *prefer*
VP → *Verb NP*
VP → *X2 PP*
X2 → *Verb NP*
VP → *Verb PP*
VP → *VP PP*
PP → *Preposition NP*

Exercise

words	Book	the	flight	trough	Houston
	S,VP, Nominal Noun,Verb [0,1]	[0,2]	S,VP,X2 [0,3]	[0,4]	S1,VP, X2 S2,VP S3 [0,5]
		Det [1,2]	NP [1,3]	[1,4]	**NP** [1,5]
			Nominal Noun [2,3]	[2,4]	Nominal [2,5]
				Preposition [3,4]	PP [3,5]
					NP, Proper-Noun [4,5]



Lexicon
Det → *that* | *this* | *the* | *a*
Noun → *book* | *flight* | *meal* | *money*
Verb → *book* | *include* | *prefer*
Pronoun → *I* | *she* | *me*
Proper-Noun → *Houston* | *NWA*
Aux → *does*
Preposition → *from* | *to* | *on* | *near* | *through*

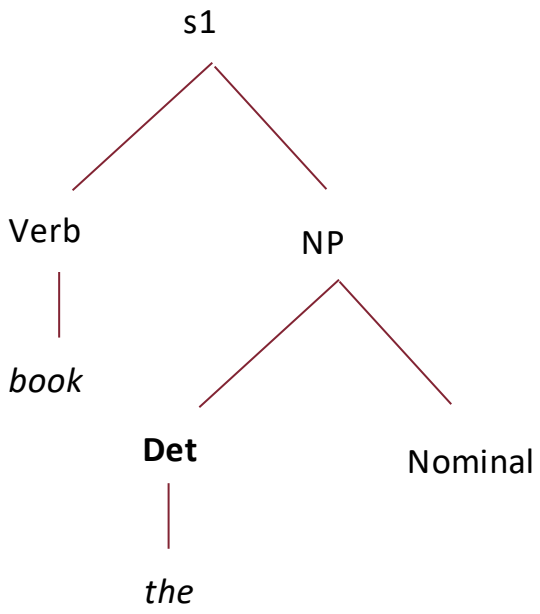
S → *NP VP*
S → *X1 VP*
X1 → *Aux NP*
S → *book* | *include* | *prefer*
S → *Verb NP*
S → *X2 PP*
S → *Verb PP*
S → *VP PP*
NP → *I* | *she* | *me*
NP → *TWA* | *Houston*
NP → *Det Nominal*
Nominal → *book* | *flight* | *meal* | *money*
Nominal → *Nominal Noun*
Nominal → *Nominal PP*
VP → *book* | *include* | *prefer*
VP → *Verb NP*
VP → *X2 PP*
X2 → *Verb NP*
VP → *Verb PP*
VP → *VP PP*
PP → *Preposition NP*

Exercise

words	Book	the	flight	trough	Houston
	S,VP, Nominal Noun,Verb [0,1]	[0,2]	S,VP,X2 [0,3]	[0,4]	S1,VP, X2 S2,VP S3 [0,5]
		Det [1,2]	NP [1,3]	[1,4]	NP [1,5]
			Nominal Noun [2,3]	[2,4]	Nominal [2,5]
				Preposition [3,4]	PP [3,5]
					NP, Proper-Noun [4,5]

Lexicon
Det → *that* | *this* | *the* | *a*
Noun → *book* | *flight* | *meal* | *money*
Verb → *book* | *include* | *prefer*
Pronoun → *I* | *she* | *me*
Proper-Noun → *Houston* | *NWA*
Aux → *does*
Preposition → *from* | *to* | *on* | *near* | *through*

S → *NP VP*
S → *X1 VP*
X1 → *Aux NP*
S → *book* | *include* | *prefer*
S → *Verb NP*
S → *X2 PP*
S → *Verb PP*
S → *VP PP*
NP → *I* | *she* | *me*
NP → *TWA* | *Houston*
NP → *Det Nominal*
Nominal → *book* | *flight* | *meal* | *money*
Nominal → *Nominal Noun*
Nominal → *Nominal PP*
VP → *book* | *include* | *prefer*
VP → *Verb NP*
VP → *X2 PP*
X2 → *Verb NP*
VP → *Verb PP*
VP → *VP PP*
PP → *Preposition NP*



Exercise

words	Book	the	flight	trough	Houston
	S,VP, Nominal Noun,Verb [0,1]	[0,2]	S,VP,X2 [0,3]	[0,4]	S1,VP, X2 S2,VP S3 [0,5]
		Det [1,2]	NP [1,3]	[1,4]	NP [1,5]
			Nominal Noun [2,3]	[2,4]	**Nominal** [2,5]
				Preposition [3,4]	PP [3,5]
					NP, Proper-Noun [4,5]

Lexicon
Det → *that* | *this* | *the* | *a*
Noun → *book* | *flight* | *meal* | *money*
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S → *NP VP*
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S → *Verb PP*
S → *VP PP*
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NP → *Det Nominal*
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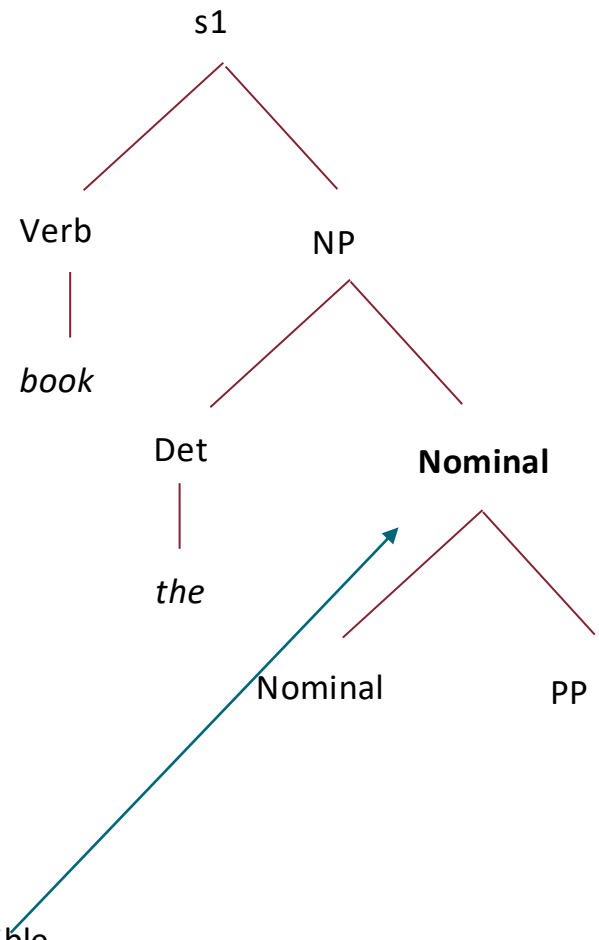
Nominal → *Nominal Noun*
Nominal → *Nominal PP*
VP → *book* | *include* | *prefer*
VP → *Verb NP*
VP → *X2 PP*
X2 → *Verb NP*
VP → *Verb PP*
VP → *VP PP*
PP → *Preposition NP*



2 possible derivations here!

Exercise

words	Book	the	flight	trough	Houston
	S,VP, Nominal Noun,Verb [0,1]	[0,2]	S,VP,X2 [0,3]	[0,4]	S1,VP, X2 S2,VP S3 [0,5]
		Det [1,2]	NP [1,3]	[1,4]	NP [1,5]
			Nominal Noun [2,3]	[2,4]	Nominal [2,5]
				Preposition [3,4]	PP [3,5]
					NP, Proper-Noun [4,5]

Lexicon
Det → *that* | *this* | *the* | *a*
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Pronoun → *I* | *she* | *me*
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S → *X1 VP*
X1 → *Aux NP*
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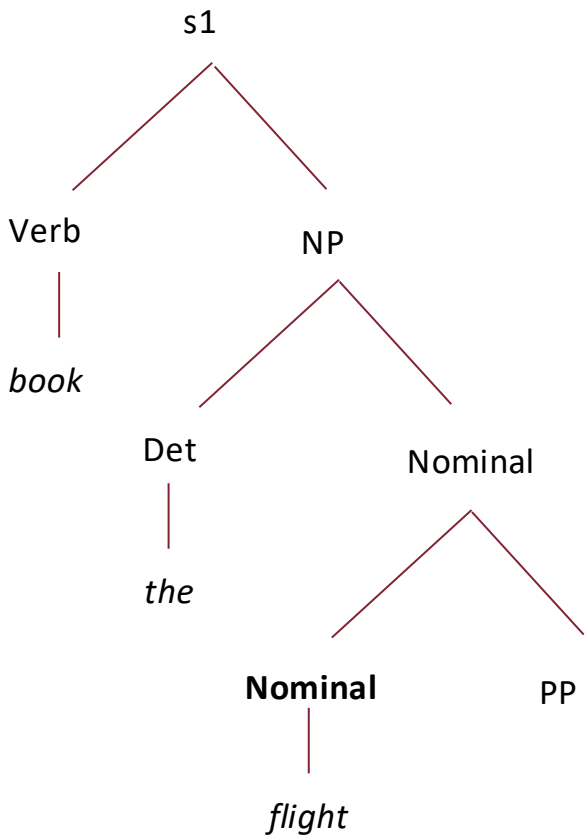
S → *Verb NP*
S → *X2 PP*
S → *Verb PP*
S → *VP PP*
NP → *I* | *she* | *me*
NP → *TWA* | *Houston*
NP → *Det Nominal*
Nominal → *book* | *flight* | *meal* | *money*
Nominal → *Nominal Noun*
Nominal → *Nominal PP*
VP → *book* | *include* | *prefer*
VP → *Verb NP*
VP → *X2 PP*
X2 → *Verb NP*
VP → *Verb PP*
VP → *VP PP*
PP → *Preposition NP*



Exercise

words	*Book*	*the*	*flight*	*trough*	*Houston*
	S,VP, Nominal Noun,Verb [0,1]	[0,2]	S,VP,X2 [0,3]	[0,4]	S1,VP, X2 S2,VP S3 [0,5]
		Det [1,2]	NP [1,3]	[1,4]	NP [1,5]
			Nominal Noun [2,3]	[2,4]	Nominal [2,5]
				Preposition [3,4]	**PP** [3,5]
					NP, Proper-Noun [4,5]

Lexicon
Det → *that* | *this* | *the* | *a*
Noun → *book* | *flight* | *meal* | *money*
Verb → *book* | *include* | *prefer*
Pronoun → *I* | *she* | *me*
Proper-Noun → *Houston* | *NWA*
Aux → *does*
Preposition → *from* | *to* | *on* | *near* | *through*

S → *NP VP*
S → *X1 VP*
X1 → *Aux NP*
S → *book* | *include* | *prefer*
S → *Verb NP*
S → *X2 PP*
S → *Verb PP*
S → *VP PP*
NP → *I* | *she* | *me*
NP → *TWA* | *Houston*
NP → *Det Nominal*
Nominal → *book* | *flight* | *meal* | *money*
Nominal → *Nominal Noun*
Nominal → *Nominal PP*
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VP → *Verb NP*
VP → *X2 PP*
X2 → *Verb NP*
VP → *Verb PP*
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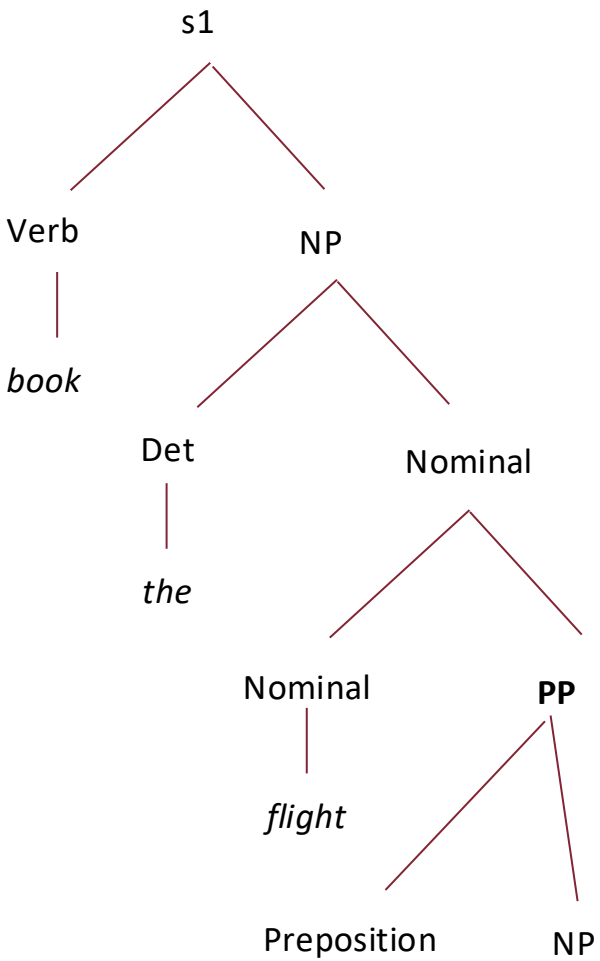

Exercise

words	Book	the	flight	trough	Houston
	S,VP, Nominal Noun,Verb [0,1]	[0,2]	S,VP,X2 [0,3]	[0,4]	S1,VP, X2 S2,VP S3 [0,5]
		Det [1,2]	NP [1,3]	[1,4]	NP [1,5]
			Nominal Noun [2,3]	[2,4]	Nominal [2,5]
				Preposition [3,4]	PP [3,5]
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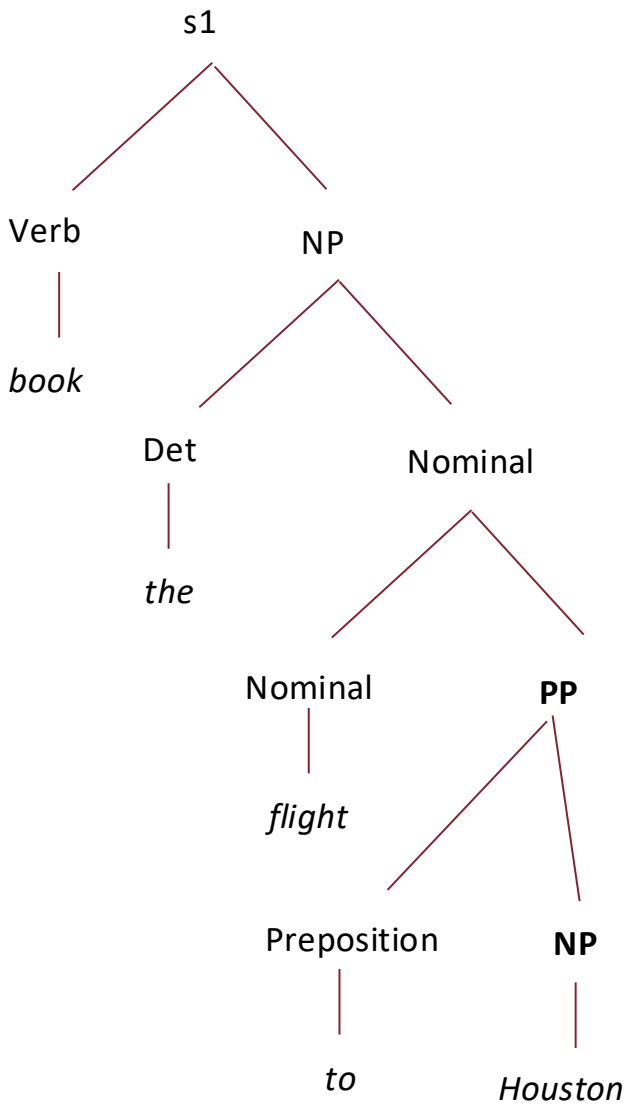

Exercise

words	Book	the	flight	trough	Houston
	S,VP, Nominal Noun,Verb [0,1]	[0,2]	S,VP,X2 [0,3]	[0,4]	S1,VP, X2 S2,VP S3 [0,5]
		Det [1,2]	NP [1,3]	[1,4]	NP [1,5]
			Nominal Noun [2,3]	[2,4]	Nominal [2,5]
				Preposition [3,4]	**PP** [3,5]
					NP, Proper-Noun [4,5]

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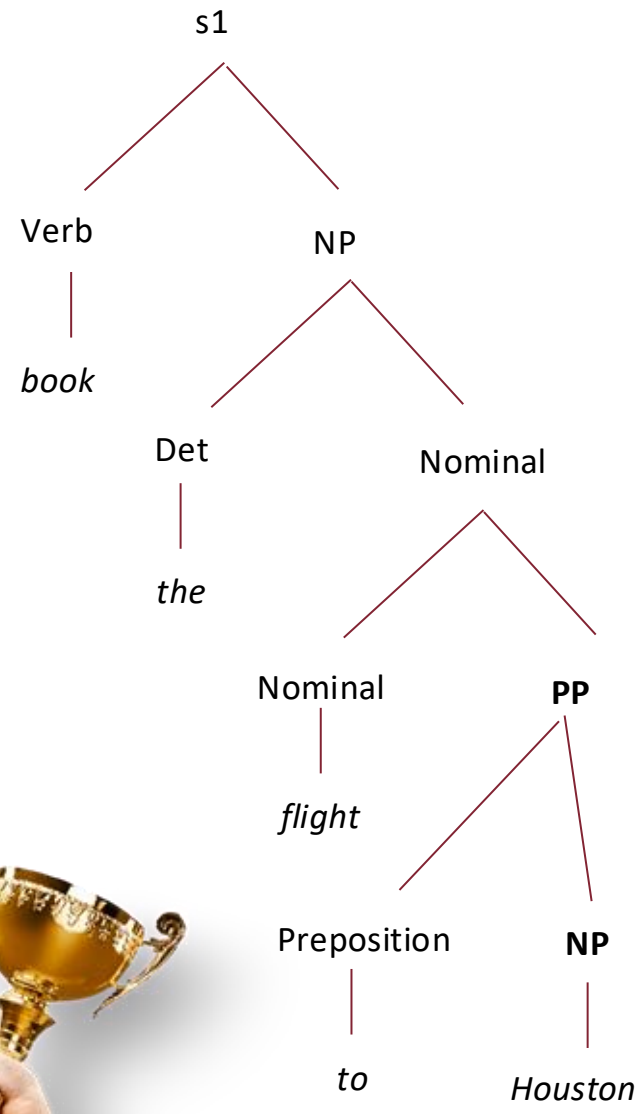
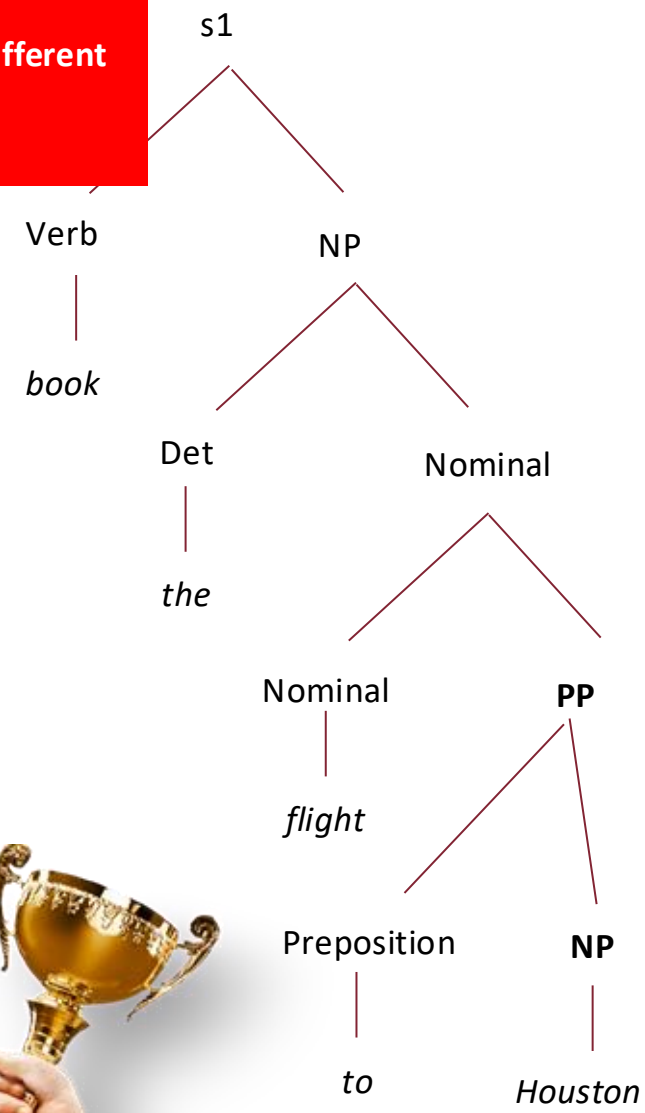


Exercise

A sentence can be recognized as part of the language, correct syntax but the parser can generate all existing different valid parses

words	*Book*	*the*			
	S, VP, Nominal Noun, Verb [0,1]	[0,2]	S, [0,		
		Det [1,2]	NP [1,3]	[1,4]	NP [1,5]
			Nominal Noun [2,3]	[2,4]	Nominal [2,5]
				Preposition [3,4]	PP [3,5]
					NP, Proper-Noun [4,5]



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X2 → *Verb NP*
VP → *Verb PP*
VP → *VP PP*
PP → *Preposition NP*

Practical Problems with CKY

Chomsky Normal Form

- We need grammars to be in CNF
  - This complicates syntax-driven approaches to semantic analysis
- Solution: keep enough information to transform the parse trees back to the original version
  - Easy for transformation rules such as: $A \rightarrow BCw \Rightarrow X \rightarrow BC$; $A \rightarrow Xw$
  - More complex for unit productions such as: $A \rightarrow B$

Exercise with CKY

- Given the following CFG grammar:

- S → NP V | V NP
- NP → Det N
- N → book | flight
- Det → the | that
- V → book | flies

- Parse the sentence “book that flight”

Probabilistic Parsing (intuition)

- The most commonly used probabilistic grammar: probabilistic CFGs
  - Each rule is assigned a probability
  - Trained on a hand-labeled Treebank grammar
  - Parsing with a probabilistic version of CKY

Probabilistic (or Stochastic) CFGs

- First proposed by Taylor Booth (1969)
- In a probabilistic CFG G = (N, T, P, S), each production

$$A \rightarrow w \; [p]$$

  is assigned a probability $p = P(w|A) = P(A \rightarrow w)$
- For each left-hand-side non-terminal A, it must hold:

$$\sum_{w} P(A \rightarrow w) = 1$$

An Example of PCFG (from Jurafsky & Martin)

$S \rightarrow NP\ VP$	[.80]
$S \rightarrow Aux\ NP\ VP$	[.15]
$S \rightarrow VP$	[.05]
$NP \rightarrow Pronoun$	[.35]
$NP \rightarrow Proper\text{-}Noun$	[.30]
$NP \rightarrow Det\ Nominal$	[.20]
$NP \rightarrow Nominal$	[.15]
$Nominal \rightarrow Noun$	[.75]
$Nominal \rightarrow Nominal\ Noun$	[.20]
$Nominal \rightarrow Nominal\ PP$	[.05]
$VP \rightarrow Verb$	[.35]
$VP \rightarrow Verb\ NP$	[.20]
$VP \rightarrow Verb\ NP\ PP$	[.10]
$VP \rightarrow Verb\ PP$	[.15]
$VP \rightarrow Verb\ NP\ NP$	[.05]
$VP \rightarrow VP\ PP$	[.15]
$PP \rightarrow Preposition\ NP$	[1.0]

$Det \rightarrow that$ [.10] | a [.30] | the [.60]	
$Noun \rightarrow book$ [.10] | $flight$ [.30]	
| $meal$ [.15] | $money$ [.05]	
| $flights$ [.40] | $dinner$ [.10]	
$Verb \rightarrow book$ [.30] | $include$ [.30]	
| $prefer$; [.40]	
$Pronoun \rightarrow I$ [.40] | she [.05]	
| me [.15] | you [.40]	
$Proper\text{-}Noun \rightarrow Houston$ [.60]	
| TWA [.40]	
$Aux \rightarrow does$ [.60] | can [40]	
$Preposition \rightarrow from$ [.30] | to [.30]	
| on [.20] | $near$ [.15]	
| $through$ [.05]	

PCFGs for Disambiguation

- So What Is the Probability of a Parse Tree?
- Defined as the product of the probabilities of all the productions used in the derivation

$$P(T,s) = \prod_{i=1}^{n} P(RHS_i \mid LHS_i)$$

where s is the sentence whose parse is T and $LHS_i \rightarrow RHS_i$ is just the i-th production in the derivation

- The joint probability P(T, s) = probability of the parse tree P(T):

$$P(T,s) = P(T)P(s \mid T) = P(T)$$

because T includes all the words in s

Probabilistic CKY Algorithm

- We just add probabilities to non-terminals in the bidimensional matrix cells
- If cell (i, j) contains non-terminal A, we specify its probability to span positions i through j of the input sentence

```
function PROBABILISTIC-CKY(words,grammar) returns most probable parse
                                                  and its probability
  for j ← from 1 to LENGTH(words) do
     for all { A | A → words[j] ∈ grammar }
       table[j − 1, j, A] ← P(A → words[j])
     for i ← from j − 2 downto 0 do
        for k ← i + 1 to j − 1 do
              for all { A | A → BC ∈ grammar,
                              and table[i,k,B] > 0 and table[k, j,C] > 0 }
                    if (table[i,j,A] < P(A → BC) × table[i,k,B] × table[k,j,C]) then
                       table[i,j,A] ← P(A → BC) × table[i,k,B] × table[k,j,C]
                       back[i,j,A] ← {k,B,C}
     return BUILD_TREE(back[1, LENGTH(words), S]), table[1, LENGTH(words), S]
```

Example of Probabilistic CKY Chart (intuition)

S	$\rightarrow$	$NP\ VP$	.80	Det	$\rightarrow$	the	.50
NP	$\rightarrow$	$Det\ N$	.30	Det	$\rightarrow$	a	.40
VP	$\rightarrow$	$V\ NP$	.20	N	$\rightarrow$	$meal$	.01
V	$\rightarrow$	$includes$	.05	N	$\rightarrow$	$flight$	.02

Det: .40 [0,1]	NP: .30 *.40 *.02 = .0024 [0,2]	[0,3]	[0,4]	[0,5]
	N: .02 [1,2]	[1,3]	[1,4]	[1,5]
		V: .05 [2,3]	[2,4]	[3,5]
			[3,4]	[3,5]
				[4,5]

To be continued…

Q&A

Resources and References

[Jurafsky&Martin, 2022] Jurafsky and Martin. Speech and Language Processing, Prentice Hall, third edition https://web.stanford.edu/~jurafsky/slp3/ed3book.pdf

**Credits

The slides of this part of the course are the result of a personal reworking of the slides and of the course material from different sources:

1. The NLP course of Prof. Roberto Navigli, Sapienza University of Rome
2. The NLP course of Prof. Simone Paolo Ponzetto, University of Mannheim, Germany
3. The NLP course of Prof. Chris Biemann, University of Hamburg, Germany
4. The NLP course of Prof. Dan Jurafsky, Stanford University, USA

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